The Semantic Ontology of Actualization

Clocks, weights, and the common architecture of relativity, thermodynamics, and quantum theory

C. Pacitani*

Abstract

Relativity, thermodynamics, and quantum theory are usually merged, when they are merged at all, by unifying their *equations*. This paper merges their *semantics* instead. We take as primitive a distinction that all three already presuppose but none formalizes: between what is actual — a finite record of events that has happened — and what is potential — a structured field of weighted alternatives that has not. An experiment is then a three-fold selection: of an intervention, of a question, and of one outcome from the weighted alternatives that question admits. Writing this as a closed recursion isolates a single unresolved object, the actualization block Σ, and everything else falls into place around it. Quantum field theory supplies the covariant law of the potential layer; the operational reconstruction programme supplies its finite-dimensional state space, and an effect-additivity postulate supplies the trace rule, so that the Born square is a theorem about weights rather than an axiom about probability. Relativity constrains Σ's covariant form, and we separate — as is not always done — the order-independence of spacelike instruments, which is standard physics, from selector-history equivariance, which is a distinct and open obligation. Thermodynamics supplies what neither supplies: irreversibility and a timescale. Two contributions are new. First, a *clock*: the period constant of any mechanical oscillator factors into a universal quadrature root $\sqrt{\pi}$ and a degree ratio, with the harmonic case uniquely self-reproducing — which is why its constant is a power of π — and the quartic case yielding the lemniscatic $G^* = \Gamma(\frac{1}{4})/\Gamma(\frac{3}{4})$. A proposition with named hypotheses states when a threshold forces the quartic, in the $A \to 0$ limit, with its finite-amplitude failure mode exhibited exactly. In a discrete model, timekeeping then splits by a decidable rigidity criterion into three exhaustive cases; the substrate natively realizes only the harmonic one, and the quartic clock is available only at an itemized price — four bodies and one additional interaction range, plus a threshold that must be actively held. Its period law is exact and its constant is recovered from simulation with no fitted scale. Second, a reinterpretation of Born weighting: in a pre-registered mechanism-level measurement, threshold statistics interpolate from amplitude weighting to occupation (Born) weighting as mode frequency exceeds the noise bandwidth, so that the Born rule appears as a *regime* of actualization rather than an axiom. We report with equal prominence the accompanying negative — that the same law did not transfer to the model's own thermal field in the sampled domain — and a purity requirement discovered along the way: the slow channel must be essentially the sole stochastic element at the threshold. We claim a shared architecture with itemized interfaces, not a unification.

*Draft, 2026-08-09. This is a foundations paper: an architecture assembled from established external results plus two declared postulates, together with a small number of exactly stated and pre-registered measurements made within one concrete discrete model. It should be cited as foundations, never as a physics result. Every quantitative claim carries its scope in the text and again in the status ledger of Section 13. All figures are computed by an accompanying script; all measured values are transcribed from locked pre-registration records listed in Appendix A.

1 Building a world, and paying for it

A physicist moving between relativity, thermodynamics, and quantum theory changes vocabulary three times. In relativity the primitive is an *event* and the structure is a partial order. In thermodynamics the primitives are *macrostates* and the structure is a monotone. In quantum theory the primitive is a *state* and the structure is an algebra of operators. The three are routinely combined at the level of equations — quantum field theory on curved backgrounds, black-hole thermodynamics, relativistic statistical mechanics — and yet the combinations remain notoriously partial, and the partiality always localizes to the same place: what happens when something *becomes the case*.

This paper proposes that the three theories already share a semantic distinction which none of them makes explicit, and that making it explicit is what the merge requires. The distinction is between the *actual* — a finite, singular record of what has occurred — and the *potential* — a structured field of weighted alternatives that have not occurred and mostly never will. Once the distinction is drawn, each theory turns out to govern a different part of it. Quantum theory, including its relativistic form, is the law of the potential layer: how weighted alternatives are structured, how they evolve, how they compose. Relativity constrains how the transition from potential to actual can be described without a preferred frame. Thermodynamics supplies the direction and the cost of that transition, and — as we will show — a quantitative condition on when it produces quantum statistics at all.

The way to test a claim like that is to try to build the thing and keep the receipts. So the paper proceeds as a construction: begin with the distinction and nothing else, and at each stage add only what the stage before it made necessary — saying, every time, what the addition cost. A section that adds nothing is a section that should not be there, and a section that adds something without naming a price is the failure mode this method exists to prevent.

Four kinds of cost recur, and the paper uses these four words for them throughout. A **postulate** is declared outright: it buys structure and is answerable only to consistency and to the reader's judgement that it was worth declaring. An **import** is an external theorem, carried in whole and conditional on its own premises, which the paper does not re-prove. A **purchase** is a piece of machinery the substrate did not supply and that has to be added to it — an interaction range, a species, a degree of freedom. And a **maintenance** is a condition that is not paid once but held: a configuration the world drifts away from unless something keeps returning it. The last is the least familiar and turns out to be the most expensive.

Two postulates are declared, in Sections 2 and 7. The imports are itemized rather than summarized, because they are the largest single thing the paper does not derive. The purchases and the maintenance are what the constructive sections are about. Section 13 totals all of it, and where that table and the body disagree, the table is right.

What this paper claims, and what it does not. It claims a *common semantic architecture with itemized interfaces* — not a unification, not a derivation of quantum mechanics, and not a new dynamical theory. The reconstruction results of Section 7 are *imported* and conditional on their own premises. The measurements of Sections 4 and 8 are made within one concrete discrete model and carry the scope of their pre-registrations. Planck's constant, the Hamiltonian, particle masses, and coupling constants are not derived anywhere in this paper. Nor is the lemniscatic constant of Section 4: it is the period of a shape the substrate is not shown to adopt, and it enters this framework by choice. The full accounting is Table 2.

Nothing has been declared yet, and nothing bought. The construction begins in the next section with the cheapest possible commitment: a single distinction, without which there is not yet anything for a physics to be about.

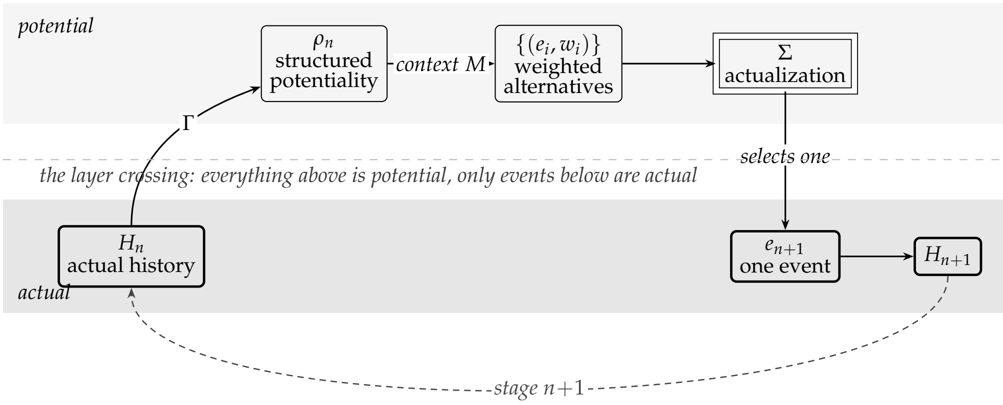

Figure 1: The shape of the recursion, drawn here before most of it has been paid for. Structure is encoded in shape and rule, never in colour: solid boxes are theorem-grade within their stated premises, thin boxes are selections or imported representations, and the double-ruled box is the paper's one open object. Actual history generates structured potentiality through a state-assignment map Γ; a chosen context resolves it into weighted alternatives; the actualization block selects one; the record extends and the cycle repeats. Everything above the lower band is potential; only the events in it are actual. *Of the objects shown, this section has bought only the two bands and the singular event.* The state-assignment map, the context and the weights are purchased in Section 7, which is where the diagram closes into the equation it depicts.

2 Primitive ontology

At stage n the **actual history** is a finite sequence of events

$$H_n = (e_1, e_2, \ldots, e_n). \tag{1}$$

There may be many possible continuations, collected in a set $\mathcal{P}(H_n) = \{e'_1, e'_2, \ldots\}$. The founding move of this architecture is to refuse to identify these two objects:

$$\text{actual history } H_n \;\neq\; \text{potential continuations } \mathcal{P}(H_n). \tag{2}$$

Postulate 1 (Singular actuality)**.** One realized experiment produces one next event e_{n+1}, so that $H_{n+1} = H_n \circ e_{n+1}$.

Postulate 1 is a declared commitment of this scaffold, not a theorem and not a consequence of any dynamics presented here. Its principal alternative — denying it, and taking all weighted continuations to be actual — is a live position in the literature and is priced symmetrically in Section 9.2. We adopt it because the architecture we are building is an architecture *of* actualization, and a theory in which nothing becomes the case has no actualization to organize.

An experiment is not merely an encounter with a probability distribution. Before any distribution is defined, two selections are made: of a controlled input $u = \Sigma_U(\mathcal{U})$ and of a measurement context $M = \Sigma_M(\mathcal{M})$. These are *operational* selections — a knob, a clock, a deterministic circuit will do — and carry no commitment about freedom. The point is logical order: *what are we doing to the system*, and *what distinction will be read out*, both precede *with what frequencies*.

One thing this buys is worth isolating before going on, because it is smaller than it looks. A record that extends has an order. What it does not have is a scale on that order, and the next section shows why that gap cannot be closed by looking at the record harder.

3 Succession is not yet time

What has been bought so far is a distinction and a rule for extending a record. That is strictly less than a world, and the gap is worth naming precisely, because it is the first place the construction has to spend.

Writing $H_{n+1} = H_n \circ e_{n+1}$ gives an *order*. An order supports two predicates, *before* and *after*, and it supports nothing else. In particular it does not support *how much*. Nothing in a growing list distinguishes a long interval from a short one, because the list has no second interval to compare against: each event is simply the next.

A duration is a ratio. To say that one interval is 3.7 of another is to have counted something that happened 3.7 times, so a unit is required before a duration is meaningful, and a unit requires a *recurrence* — a feature that comes back, so that returns can be counted. Monotone succession never returns. Whatever supplies the recurrence is therefore not the index; it is a physical thing inside the history, and it must be paid for.

Two consequences follow, and they set up everything after this section.

First, the demand has a definite shape: a bounded degree of freedom, a law that restores it, and a constant relating how far it swings to how long it takes. That constant is where the price will show up, because the restoring law is not free — a substrate either supplies one or does not.

Second, and less obviously, the same argument applies to the paper's own bookkeeping. The recursion this architecture will run on is indexed by a stage, and if the index is to mean elapsed time rather than list position, something physical has to carry it. A clock is therefore not a component of the architecture among others; it is a precondition of the architecture's own description of itself. That thread is picked up in Section 10, once there is a recursion on the page to be indexed.

So: what is the cheapest recurrence a discrete substrate can host, what constant does it keep, and what does it cost?

4 Time: the clock

Four questions, in the order Section 3 leaves them. What does a period cost, in the sense of what fixes its constant? What shapes can a substrate make available? Which of them, if any, is forced? And what does the shortfall between those two answers cost, and buy? Each subsection below answers exactly one of them.

4.1 The setting

The model is a discrete one: point bodies on a lattice, interacting through a compact central pair potential with a single minimum at separation r_0, and carrying a polarity that makes the interaction graph bipartite. Two features matter. Every bond in an equilibrium sits *at* its potential minimum, so the configuration carries zero actual tension; and the potential has compact support, so "bonded" is a sharp predicate. What follows is exact within that model.

4.2 What a period costs

Before asking which oscillator a substrate supplies, it is worth asking what any of them charges. Take a symmetric confining potential $V = \lambda |x|^n$ and release a body of mass m from rest at amplitude A, so $E = \lambda A^n$. Energy conservation fixes the speed, $v = \sqrt{2(E-V)/m}$, and a quarter period is the time to cross from the centre to the turning point:

$$\frac{T}{4} = \int_0^A \frac{\mathrm{d}x}{v} = A\sqrt{\frac{m}{2E}}\, I(n), \qquad I(n) = \int_0^1 \frac{\mathrm{d}u}{\sqrt{1-u^n}} = \frac{1}{n} B\left(\tfrac{1}{n}, \tfrac{1}{2}\right). \tag{3}$$

Everything specific to the clock is in the $I(n)$ of (3), and it factors — in a way that says which part of a clock constant is kinematics and which part is the potential:

$$I(n) = \underbrace{\sqrt{\pi}}_{\text{quadrature root}} \cdot \underbrace{\frac{\Gamma(1/n)}{n\,\Gamma(1/n+1/2)}}_{\text{degree ratio}}. \tag{4}$$

The first factor is $\Gamma(\frac{1}{2})$, and it is there because the integrand carries a square root — speed is the root of an energy. That is true of *every* mechanical clock whatever its potential, so $\sqrt{\pi}$ is a charge nobody escapes. The second factor is the only thing that moves when the potential's degree does.

n	$I(n)$	degree ratio	
2	1.5707963268	0.8862269255	$=\sqrt{\pi}/2$
3	1.4021821053	0.7910965381	
4	1.3110287771	0.7396687798	$=G^*/4$
5	1.2537306248	0.7073417591	
6	1.2143253239	0.6851096988	
8	1.1635925712	0.6564868082	

Two rows are special, and their relation is the point.

At $n=2$ the degree ratio is $\Gamma(\frac{1}{2})/\Gamma(1)=\sqrt{\pi}/2$ — *the quadrature root again*. The harmonic case is the unique self-reproducing one, so its constant is $I(2)=\sqrt{\pi}\cdot\sqrt{\pi}/2=\pi/2$: a power of π and nothing else. This is why the circle constant turns up in timekeeping at all, and it is a statement about the *root*, not about circles.

At $n=4$ the degree ratio is $G^*/4$, giving $I(4)=\sqrt{\pi}\,G^*/4=\varpi/2$, half the lemniscate constant. So π and G^* are not rival constants competing to be fundamental. They are the same kinematic factor multiplied by two different degree ratios, and $G^*=2\varpi/\sqrt{\pi}$ is that statement rearranged.

4.3 A decidable trichotomy

Let R be the rigidity matrix of a framework, whose rows are the stretch rates $\hat{u}_e\cdot(q_i-q_j)$ of each bond under a displacement field q. Expanding a path $p(d)=p_0+d\,q+d^2w$, the stretch of bond e is $d\,(Rq)_e+d^2[\kappa_e(q)+(Rw)_e]+O(d^3)$ with $\kappa_e(q)=|q_i-q_j|_\perp^2/2\ell_e$. Three cases are exhaustive.

Criterion 1 (The rigidity trichotomy)**.** *At zero tension, along a direction q:*

1. *if $Rq\neq 0$, the direction is first-order rigid and the energy grows as d^2: $n=2$, a harmonic clock;*
2. *if $Rq=0$ and $\kappa(q)\perp$ coker R, the flex extends to second order and generically to a finite mechanism: $n\geq 6$ or $n=\infty$;*
3. *if $Rq=0$ but $\langle\omega,\kappa(q)\rangle\neq 0$ for a self-stress $\omega\in$ coker R, the flex is* blocked*, and minimizing over w gives exactly*

$$E(d)=\lambda(q)\,d^4+O(d^6),\qquad \lambda(q)=\frac{\langle\hat{\omega},\kappa(q)\rangle^2}{2\sum_e\hat{\omega}_e^2/k_e}, \tag{5}$$

so $n=4$.

Hence $n=4$ at zero tension exists if and only if *a self-stress exists and blocks the flex.*

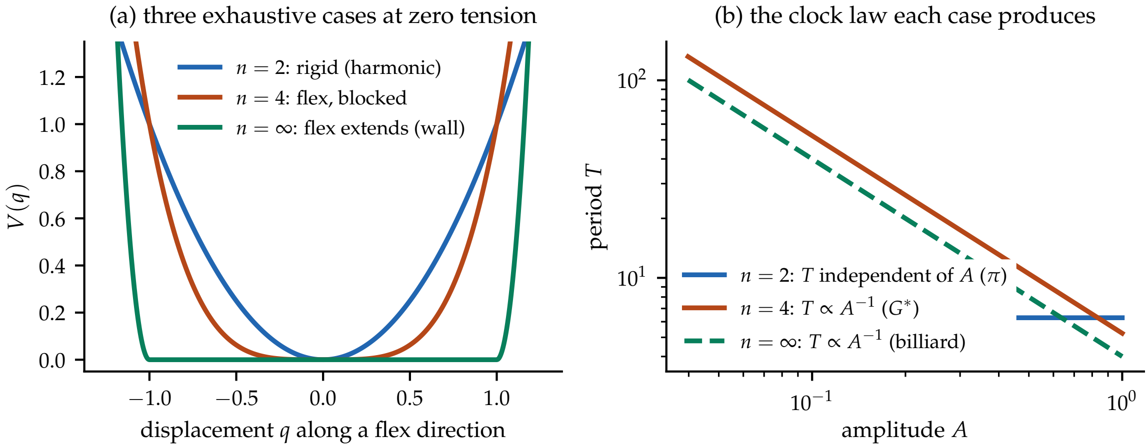

Figure 2: The rigidity trichotomy and the clocks it produces. (a) The three exhaustive cases of Criterion 1 at zero tension. (b) The corresponding period laws: the harmonic case is amplitude-independent with constant π; the quartic case has $T \propto A^{-1}$ with constant G^*; a hard wall also gives $T \propto A^{-1}$ but with no special constant.

Two corollaries follow at once: no self-stress means no $n = 4$, and actual tension means $n = 2$, since a nonzero equilibrium stress puts a d^2 term on any flex it blocks. The quartic case lives only at the degenerate point — zero actual tension with nonzero *possible* stress. This makes what is otherwise a question about potential shapes decidable by linear algebra.

The trichotomy matters because the three cases keep time differently (Figure 2). The harmonic case is isochronous and its period constant is π, which (4) has already explained. The quartic case is the anti-pendulum: its frequency grows with amplitude, and its period constant is the lemniscatic

$$G^* = \frac{\Gamma(\frac{1}{4})}{\Gamma(\frac{3}{4})} = 2.958675119188639\ldots \tag{6}$$

4.4 What forces the quartic, and what does not

The trichotomy says a quartic clock is *available*. A stronger statement is available too, and because it is the one most likely to be over-read, it is stated with its hypotheses attached and its failure modes named.[1]

Proposition 1 (Threshold forcing)**.** *Let a single degree of freedom evolve in a potential $V(x;\mu)$ such that*

(H1) *V is smooth in x and in a control parameter μ, with one degree of freedom;*

(H2) *V is invariant under $x \mapsto -x$;*

(H3) *the family crosses $\mu = 0$ transversally, at which point $\partial_x^2 V(0;0) = 0$;*

(H4) *the quartic coefficient is nonzero, $\lambda \equiv \frac{1}{4!}\partial_x^4 V(0;0) > 0$.*

Then at $\mu = 0$ the leading behaviour is $V = \lambda x^4 + O(x^6)$, and in the limit $A \to 0$

$$T \cdot A \;=\; \sqrt{\pi}\, G^* \sqrt{m/2\lambda}.$$

[1] The reading of the quartic as the clock of marginal stability, together with the period law and the anti-pendulum framing, originates in a companion manuscript on the edge-of-stability clock; what is added here is the statement as a proposition with named hypotheses, the factorisation (4), and the finite-amplitude term.

The content is in what each hypothesis excludes. (H2) is what kills the cubic — and it kills it *by symmetry, not by tuning*, which is why the quartic is generic here rather than fine-tuned. (H3) is the threshold. (H4) is nondegeneracy. And the $A \to 0$ clause is not decoration: it is the whole of the second failure mode below.

The substrate is not known to sit at a threshold. Proposition 1 forces the quartic *given* a threshold; it says nothing about why a substrate would be at one. Since $\mu = 0$ is a codimension-one surface, the generic configuration is not on it, and staying on it is a condition that must be actively held. The price is therefore a maintenance cost paid continuously, not a construction cost paid once — which is the sense in which this is a clock at the *edge* of stability rather than a clock in a well.

Two ways it fails, and only one of them is obvious. Dropping (H4) entirely gives a pure sextic and a single different constant, $4I(6)/\sqrt{\pi} = 2.7404387952$. That failure is visible.

The other is not, and it is a genuine counterexample to any finite-amplitude version of the proposition. Take $V = \lambda x^4 + \nu x^6$ with both terms present, and write $r = \nu A^2/\lambda$. Then

$$T \cdot A = 4\sqrt{\frac{m}{2\lambda}} \int_0^1 \frac{\mathrm{d}u}{\sqrt{(1-u^4) + r\,(1-u^6)}}, \tag{7}$$

and the constant sweeps *continuously* with r: the bracket runs 5.2441, 5.2133, 4.9589, 3.5563 at $r = 0$, 0.01, 0.1, 1. Every member of this family has $\lambda \neq 0$, so (H4) does not exclude it. What excludes it is the limit: $r \propto A^2$, so the drift vanishes with the amplitude and G^* is the $A \to 0$ endpoint, not a finite-amplitude value.

This is worth stating plainly because the same term appears in this paper's own measurement. The twelve-degree-of-freedom simulation of Section 4.7 holds $T \cdot A$ constant only to 0.33% across a sixfold amplitude range, and recovers G^* to 2×10^{-6} only in the $A \to 0$ extrapolation. A 0.33% deficit is $r = 0.0056$. The residual drift and the counterexample family are one term seen twice.

4.5 Off the threshold, exactly

The proposition holds *at* $\mu = 0$. Since the whole price is that $\mu = 0$ must be held, what happens as it slips is not a footnote — it is the thing being paid for, and it is exactly solvable.

For the normal form $V = \frac{1}{2}\mu x^2 + \lambda x^4$ the quadrature closes in elliptic functions: the orbit is $x(t) = A\,\mathrm{cn}(\Omega t; \kappa)$ with $\Omega^2 = (\mu + 4\lambda A^2)/m$, and writing everything in terms of the modulus

$$\kappa^2 = \frac{2\lambda A^2}{\mu + 4\lambda A^2} \tag{8}$$

eliminates μ and A separately, leaving a single universal curve:

$$\boxed{T\,A\sqrt{\tfrac{2\lambda}{m}} \;=\; 4\,\kappa\,K(\kappa)} \tag{9}$$

with K the complete elliptic integral of the first kind. Every oscillation at every detuning lies on this one curve: the harmonic side at $\kappa^2 \to 0$, the double-well side at $\frac{1}{2} < \kappa^2 < 1$, and the quartic clock at exactly $\kappa^2 = \frac{1}{2}$, where $4\kappa K = \sqrt{\pi}\,G^*$ and (10) is recovered.

Two things follow, and the second is the more interesting.

The threshold is a self-dual point. $\kappa = 1/\sqrt{2}$ is the fixed point of the exchange $\kappa \leftrightarrow \kappa'$ between a modulus and its complement, where $K' = K$. So G^* is not merely the constant of one potential among many; it is the value the period takes at the one modulus that is its own

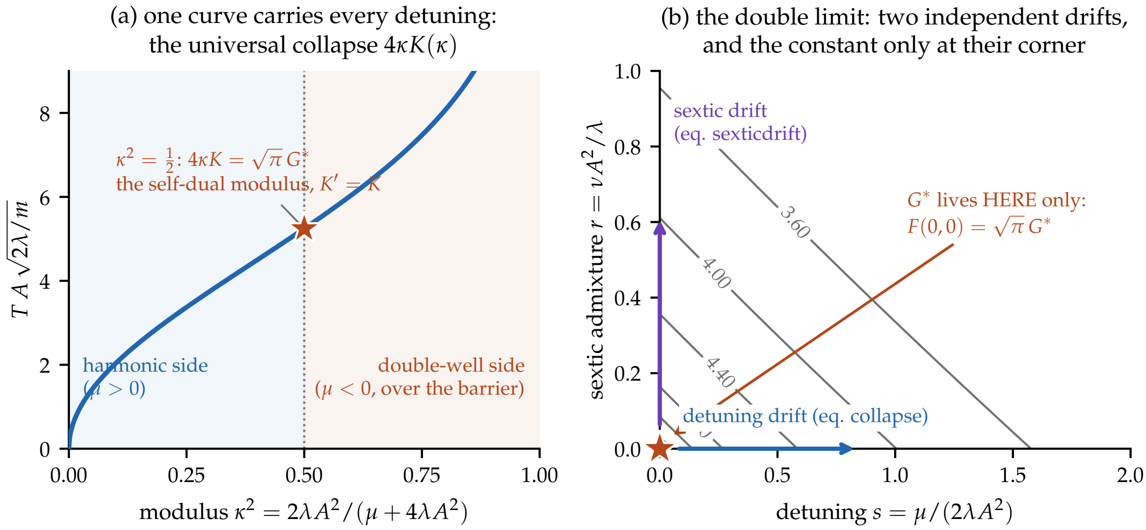

Figure 3: The crossover, exactly, and the double limit. (a) Every oscillation of the normal form at every detuning lies on the single curve $TA\sqrt{2\lambda/m} = 4\kappa K(\kappa)$: harmonic side as $\kappa^2 \to 0$, oscillation over the double-well barrier for $\kappa^2 > \frac{1}{2}$, and the quartic clock at exactly $\kappa^2 = \frac{1}{2}$ — the self-dual modulus $K' = K$ — where the value is $\sqrt{\pi}\, G^* = 5.2441$. (b) The two drift directions are independent: detuning s moves the constant along (9), sextic admixture r moves it along (7), and G^* is attained only at the corner $(0,0)$ — the double limit $\mu \to 0$, $A \to 0$. Contours are exact quadrature; the corner value agrees with $\sqrt{\pi}\, G^*$ to 10^{-9}.

complement. That is the same fact (4) reports as $n = 2$ being self-reproducing, seen through the elliptic parametrisation rather than the Gamma one.

The constant drifts two ways, and both are now exact. Detuning moves κ^2 off $\frac{1}{2}$ along (9); a sextic admixture moves the constant along (7) in r. They are independent perturbations — one changes the quadratic coefficient, the other adds a higher one — and G^* is the *double* limit $\mu \to 0$, $A \to 0$. Neither limit alone suffices, which is a sharper statement of the price than either drift gives on its own. Figure 3 draws both statements at once: the one curve every detuning lies on, and the two-drift plane whose corner is the only point where the constant lives.

4.6 The native case is harmonic

Which cases does the model realize by itself? Its zero-tension equilibria are exactly the frameworks whose active bonds all sit at r_0 with opposite-polarity endpoints — unit-edge bipartite frameworks. A screen over every connected minimum-degree-two bipartite interaction graph up to $N = 7$, plus the cube graph, found **no self-stressed embedding**: seventeen graph classes, zero hits, with constraint-clean minima of $\lambda_{\min}(RR^\mathsf{T})$ bounded away from zero at 1.03×10^{-2}. Exactly, the cubic checkerboard blocks at $L = 2, 3, 4$ have self-stress dimension zero, so all their flexes extend. Larger stressed objects contract to networks of straight unit chains meeting at joints, and the two minimal joint graphs are squeezed by geometry: an interior four-joint configuration has a joint angle of at most 30° against a clearance requirement of 28.96°, and an exhaustive integer-distance search — 153,898 integral quadruples to diameter 60, 7,177 of them with an interior point — tops out at 26.14°.

One object survives the static gates: a nineteen-body hexagonal wheel, all twenty-four bonds exactly unit, with a one-dimensional self-stress (rim in tension, spokes in compression) and every individual chain-bowing flex blocked. It is, to our knowledge, the first zero-tension self-stressed equilibrium exhibited in this model. *And it fails as a clock.* Its quartic form is a squared difference, $E_4 \propto (\sum \kappa_{\text{rim}} - \sum \kappa_{\text{spoke}})^2$, so the combined flex that bows the rim and buckles the spokes equally costs nothing: the blocking form on the 28-dimensional nontrivial flex space is sign-indefinite, with eigenvalues from -1.467 to $+0.743$. Direct integration confirms it, the constrained static energy collapsing to $\sim 10^{-12}$. The verified per-chain quartic was a chord across a flat valley.

The failure sharpens the criterion, and the sharpened form is a genuine result of this paper:

Criterion 2 (Buckling)**.** *Chain stations are free hinges, so a* compressed *chain of length ≥ 2 buckles at zero energy; only tensioned chains are straight-stable. A native $n = 4$ clock therefore requires, beyond Criterion 1:* (1) *a self-stress;* (2) *a blocking form definite on the* whole *flex space, not merely mode by mode; and* (3) *every compressed member a* single bond.

Because an all-tension self-stress is impossible on a finite framework, condition (3) forces mixed structures — single-bond struts together with straight tensioned chains. Whether such an integer "unit-strut tensegrity" exists in this model is open; the first candidate family is killed exactly, its closure condition $s^2 - 4k^2 = -1$ being impossible modulo 4.

4.7 The minimum viable clock

If the quartic clock is not native, what is the cheapest object that realizes it? The answer is exact. Take four bodies $A^+B^-C^+D^-$ collinear at positions $0, 1, 2, 3$, with the three unit bonds at their minimum (curvature k_1) and *one additional interaction species* closing A to D at its own minimum at range $3r_0$ (curvature k_2). All four bonds are at zero tension. Then rank $R = 3$, the self-stress is one-dimensional with $\omega = (1, 1, 1, -1)$, and the blocking form on the four-dimensional nontrivial flex space has eigenvalues $\{2,\ 10/3\}$ with kernel exactly the trivial motions — positive definite on *every* nontrivial flex. The chain is prestress-stable: $n = 4$ in all directions. Crucially its one compressed member is a single bond, so Criterion 2 is satisfied where the wheel failed.

The mirror-even zero-momentum mode $q = (-1, +1, +1, -1)U$ lies on a dynamically invariant manifold, so it closes exactly as a one-degree-of-freedom quartic oscillator — an instance of (5), with the blocked-flex coefficient evaluated on this stress — with

$$\lambda_{\text{eff}} = \frac{8k_1k_2}{k_1 + 3k_2}, \qquad m_{\text{eff}} = 4, \qquad \boxed{T \cdot A = \sqrt{\pi}\, G^* \sqrt{\frac{m_{\text{eff}}}{2\lambda_{\text{eff}}}} = \sqrt{\pi}\, G^* \, \frac{\sqrt{k_1 + 3k_2}}{2\sqrt{k_1k_2}}} \tag{10}$$

Direct integration of the reduced mode reproduces (10) to machine precision, and a full twelve-degree-of-freedom conservative simulation of the four-body system holds $T \cdot A$ constant to 0.33% across a sixfold amplitude range, recovering G^* to 2×10^{-6} in the $A \to 0$ extrapolation *with no fitted scale* (Figure 4).

Minimality is exact by enumeration: at $N \leq 3$ the opposite-polarity graphs are trees and cannot carry stress; at $N = 4$ the single-scale embeddings have no stressed realization; and polarity alternation forces the closure span to be odd, hence $3r_0$ at minimum. **Four bodies and one additional interaction range is the floor.**

Two remarks about $\omega = (1, 1, 1, -1)$, since the sign pattern invites a reading it will not support. It is indexed by *bonds*, not directions, and its signs are tension and compression: the lone minus sits on the closure bond, forced both by the impossibility of an all-tension self-stress on a finite framework and by a four-cycle having exactly one closure. The object that could carry a signature is the blocking form, whose coefficients these are — and it is positive *semi*-definite, spectrum $\{0, 0, 2, 10/3\}$, with no negative direction at all. Written in the consecutive stretches u_i it is $\propto 3\sum u_i^2 - (\sum u_i)^2$, the variance: the minus subtracts a mean and thereby removes the rigid-motion zero mode. It does not open a cone. An *indefinite* blocking form is precisely a buckling mode, which is how the wheel of Section 4.6 failed.

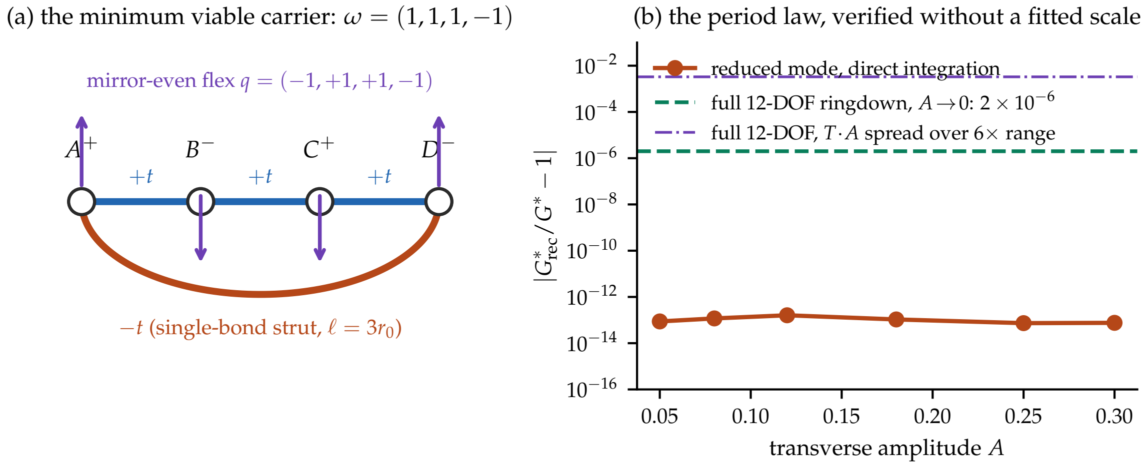

Figure 4: The minimum viable clock carrier. (a) Four bodies, three unit bonds carrying tension and one single-bond strut at range $3r_0$ carrying compression; the self-stress is $\omega = (1, 1, 1, -1)$ and the blocking form is positive definite on every nontrivial flex, so the object is prestress-stable at zero tension. (b) The period law (10) verified without a fitted scale: direct integration of the reduced mode reproduces it to machine precision, and the two horizontal lines mark the independent full twelve-degree-of-freedom result.

And the carrier cannot test what it was built beside. Recorded here, at the point of purchase, rather than where it is later needed: this clock is a mechanical framework, $m\ddot{q} = -\nabla V(|q_i - q_j|)$, and such an equation is *Galilean* invariant. Boosting it leaves the period exactly unchanged at any velocity, so it would report zero time dilation whatever the substrate did. The carrier is therefore disqualified as an instrument for the relativistic obligations of Section 9 — not by a defect in its construction, but by the kind of object it is.

The price. One additional interaction species, plus its scale parameters. What it buys is the quartic clock exactly. What it does not buy is a spectral window: with $k_1 = k_2 = \varepsilon$, clearing the propagating band requires $\varepsilon A_{\max}^2 > 1.0556$ against the field band top $2 \arcsin(1/\sqrt{3}) = 1.230959$, i.e. $\varepsilon > 4.22$ at $A_{\max} = 0.5$, far above any value the model otherwise motivates. The honest reading is that the carrier's true price is matter stiffness at field-stiffness parity. Its falsifier is stated in Appendix A.

4.8 What the lemniscatic price buys

The price is now fully stated: one interaction species, and a threshold that must be held. What has not been stated is what it returns. A ledger that records a cost and not its purchase is an unstable thing to leave standing, and the structure of the last three subsections settles it.

Read the table of (4) again, this time as a list of degeneracies. Only $n = 2$ has nonzero second-order stiffness; every $n \geq 4$ is second-order *free* and still oscillates. Ordinarily those properties exclude one another — a stiff mode oscillates but resists being moved, a soft mode moves freely but does not oscillate — and the quartic is the *minimal* escape, needing exactly one vanishing Taylor coefficient where $n = 6$ needs two. G^* is the period constant of that minimal degeneracy.

The purchase is then exact. Fixing mass, displacement and period, the energy to hold the mode displaced is

$$\frac{E\,T^2}{mA^2} = 2\pi^2 \ \ \text{(harmonic)}, \qquad \frac{\pi}{2} G^{*2} \ \ \text{(quartic)}, \qquad \frac{E_{\text{quartic}}}{E_{\text{harmonic}}} = \frac{G^{*2}}{4\pi} = 0.6966, \tag{11}$$

the stiffness cancelling from both, so the ratio is pure G^*. And at displacement a the ratio is

$(a/A)^2 G^{*2}/4\pi$, which vanishes: a degenerate direction is free to first order, so at a tenth of the reference displacement the quartic mode is 144 times cheaper to move. Figure 5 sets the two clocks beside each other — their constants as arc lengths, their orbits at equal energy, and the threshold surface on which the quartic one lives.

Proposition 1 located the genericity, and it is worth saying what that buys as well as what it costs. The quartic clock is the normal form of a perfect Euler column at critical load and of any mean-field symmetry-breaking order parameter; what is non-generic about it is not the shape of the potential but the *location*.

Why the maintenance is expensive, and not merely required. There is a sharper reason than codimension, and it is the same fact that made horology choose $n = 2$. A clock whose frequency tracks its own amplitude converts every amplitude perturbation into a frequency perturbation. Isochronism is not an aesthetic preference but *noise immunity*: the pendulum's $\partial T/\partial A \approx 0$ is the whole reason clocks work, and the anti-pendulum maximises exactly the coupling the pendulum was chosen to suppress. Near threshold the mode is soft, susceptibilities grow and relaxation slows — so the very sensitivity that makes critical points interesting makes them poor timekeepers.

That inverts usefully. A defect of a clock is a virtue of a sensor, and what this oscillator's timing is exquisitely sensitive to is *its own distance from the threshold*: by (9) the measured (T, A) pair fixes κ, and by (8) κ is the order parameter of the detuning. So a marginal clock reads out the quantity it must hold fixed. Whether that is a mechanism for holding it, or merely the statement that the error signal exists, is exactly the open problem of Section 14. Figure 6 draws the loop with its one missing edge.

So the price is a maintenance cost, continuously paid, not a construction cost; and (11) is what it returns. *π is native because sitting off the threshold is generic; G^* is priced because sitting on it must be actively held* — and what the holding buys is that timekeeping and steerability stop trading against each other. A carrier required only to be stable will be harmonic, since stability alone selects a point off the threshold; a carrier that must also be reconfigurable at low cost is pushed onto it.

This is also where the agency reading acquires a mechanism rather than an analogy. An agent, thermodynamically, is a system that spends free energy to hold a configuration against drift — and holding $\mu = 0$ is exactly that: an unmaintained system falls off the threshold and becomes a harmonic clock. The sharp form of the question is therefore not whether agents keep lemniscatic time, but that *a G^* clock is one which must be actively maintained, and what the maintenance buys is cheap actuation*. That is a claim about control cost. It neither requires nor supplies anything about consciousness, and marginal stability — a genuine hallmark of adaptive systems — is equally a property of sandpiles and buckling columns.

This interprets G^*; it does not recover it. The proposition forces the quartic *given* a threshold, a reflection symmetry and a nonvanishing quartic coefficient. Nothing here forces a substrate to sit at a threshold, and **no valid carrier has produced** G^*: the native single-scale screen returned zero, the four-body carrier that does produce it needs a purchased interaction species and is Galilean and therefore disqualified as a relativistic instrument, and the one carrier in this paper that demonstrably dilates is isochronous — a π-clock. G^* enters this framework by choice. Nor does the argument reach agency: it is a selection statement about efficient design, and marginal stability, while a hallmark of adaptive systems, is equally a property of sandpiles and buckling columns.

4.9 What a clock is owed by the rest of the paper

One forward debt is worth naming before leaving. The recursion this architecture runs on is indexed by a stage, and an index is not a duration: something physical has to carry it. That

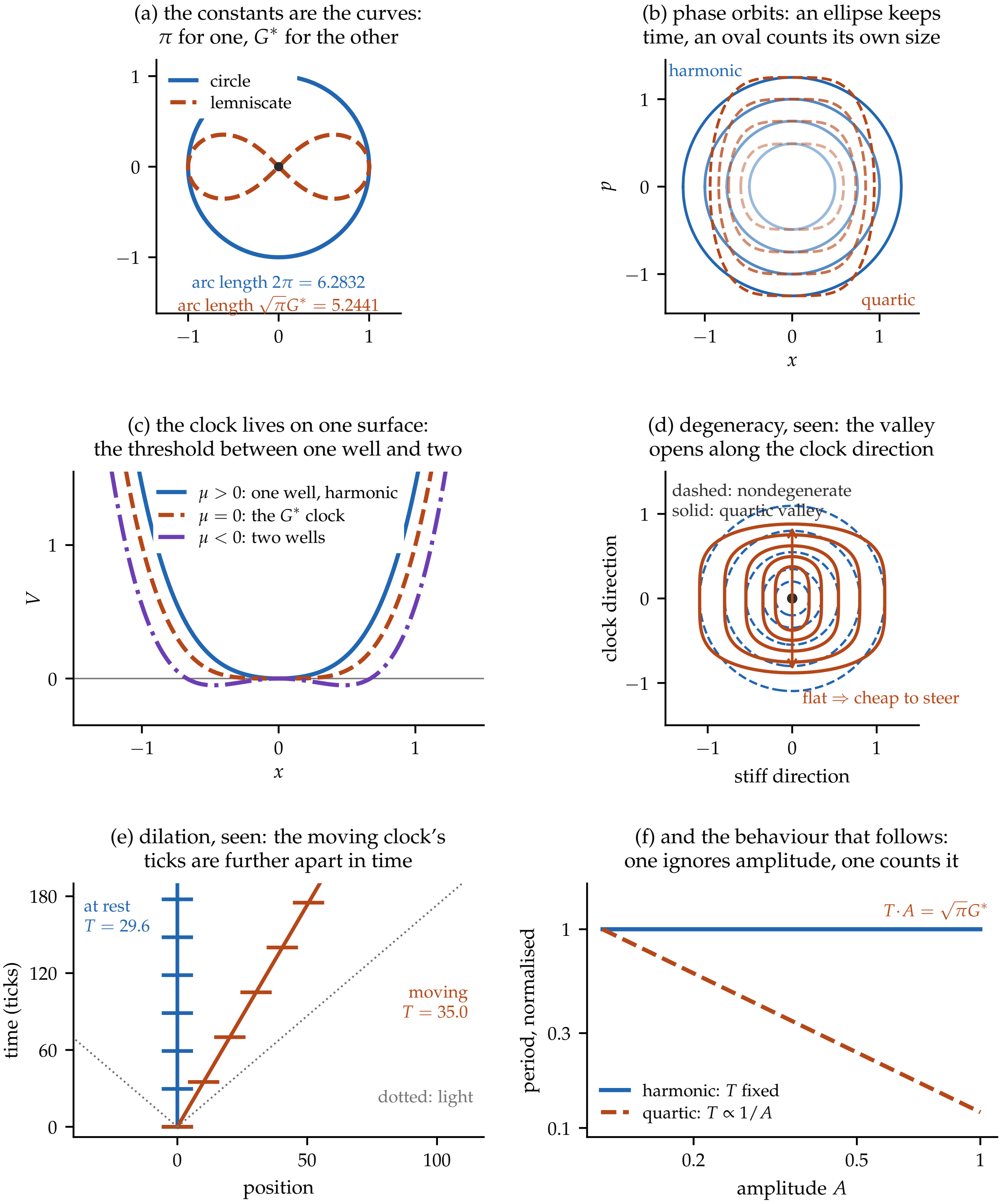

Figure 5: The two clocks, seen. (a) The constants are not normalisations but arc lengths: the unit circle's is $2\pi = 6.2832$, the unit lemniscate's is $\sqrt{\pi}G^* = 5.2441$ — the latter integrated numerically here and agreeing with $B(\frac{1}{4}, \frac{1}{2})$ to 10^{-7}. (b) Phase orbits at equal energy steps: harmonic level sets are ellipses traversed uniformly, so the period ignores the orbit's size; quartic ones are flat-flanked ovals and are traversed faster as they grow. (c) The degenerate clock does not live at a generic point but on a single surface — the threshold $\mu = 0$ between one well and two, where the cubic is forbidden by symmetry and the quartic is what remains. This is the picture of Proposition 1: the codimension-one surface is the hypothesis, and being off it is the generic case. (d) The degeneracy made visible: contours about a nondegenerate minimum are closed ellipses (dashed), while the quartic valley (solid) opens along the clock direction. That flatness is the cheapness of (11). (e) Dilation as geometry, drawn from the measured kink periods: tick events on two worldlines, one at rest and one at $u = 0.5\,C$, with the moving clock's ticks further apart in time. (f) The behaviour that follows.

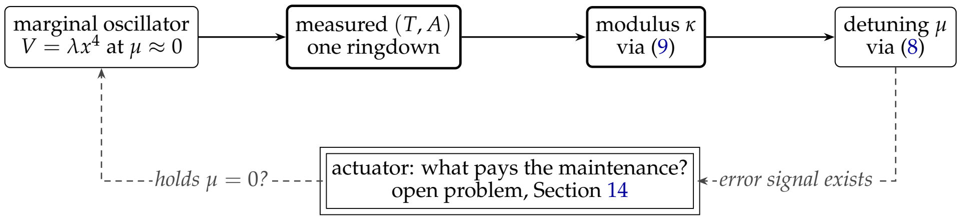

Figure 6: The sensor inversion, and the missing edge. A marginal clock's timing fixes its own distance from the threshold — the solid path is exact — so the error signal needed to hold $\mu = 0$ demonstrably exists. What would consume it is the double-ruled block: nothing in this paper constructs an actuator, and that absence is the open problem the maintenance cost points at.

makes the clock not a subplot of the architecture but a precondition of its own bookkeeping. Section 10 pays this out, once there is a recursion on the page to be indexed.

A clock, though, is not yet a record. One that is reset by every fluctuation counts nothing, because counting requires that the last tick still be there when the next arrives. Persistence is a separate purchase and it is made in the same currency — an energy barrier — which is the subject of the next section.

5 Memory: the register

A clock alone carries no past. The minimal memory in the same model is a single body cradled by three anchors on an equilateral triangle of side $s < \sqrt{3}$: for such s there are two mirror equilibria at heights $\pm h$, $h = \sqrt{1 - s^2/3}$, with all three bonds at exactly r_0.

The barrier between them is not a fitted quantity. A watershed computation over the *entire* configuration space — union-find over energy-sorted cells, so that no cheaper channel can hide — gives a barrier of 1.016–1.021 ε across $s \in [0.9, 1.3]$, against a hinge path (break one bond, swing on the remaining two) of exactly 1.000 ε; the residual few per cent is grid resolution and the third-bond tail. The direct route through the centre costs about 30ε. So

Proposition 2 (Register barrier)**.** *The register's barrier is one bond's dissociation depth, essentially independent of the geometry:* $\Delta E = \varepsilon$*.*

Two structural observations follow, and the first is the reason this section sits beside the last. The register must be *rigid with a barrier*; the clock must be *flexible but blocked*. They occupy opposite corners of the very same rigidity criterion. And the same constant ε that sets the bond depth sets the retention time, through the Arrhenius form $\tau_{\text{flip}} \sim \nu_0^{-1} e^{\varepsilon/T}$ under any declared noise ensemble (Figure 7). One constant prices binding, clock rate, and memory at once.

Both purchases so far were quoted in the same unit, ε, and neither section said what fixes it or what it costs to keep paying. That is a thermodynamic question — a barrier is only a barrier relative to a temperature — and it is the subject of the next section.

6 Thermodynamics: irreversibility and timescale

Thermodynamics enters this architecture at three places, and in each it supplies something neither the weight layer nor the relativistic structure can.

Actualization is irreversible, and the model has it natively. A reversible evolution preserves information; the transition from potential to actual does not. In the concrete model this is not an interpretive overlay but a measured property of the dynamics: the manifestation event is a threshold projection, and a single such act removes about 37% of the field norm. The inversion

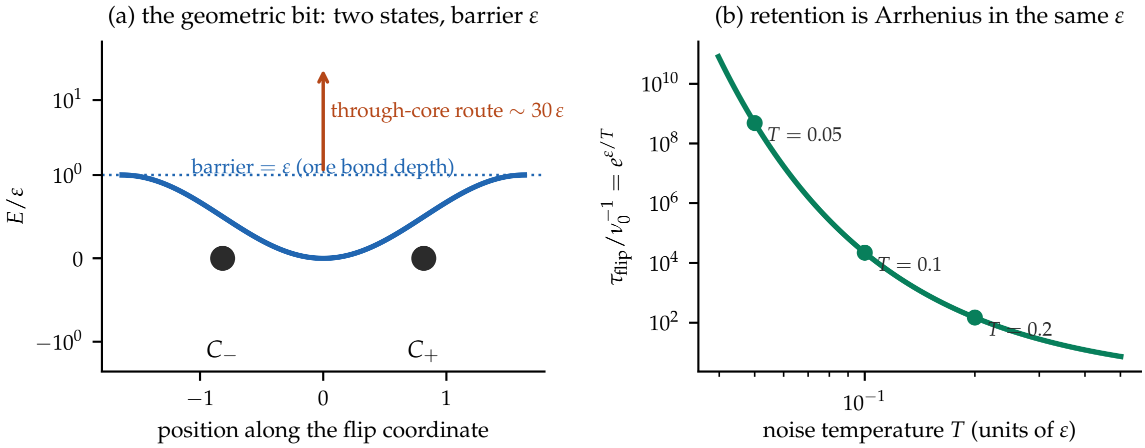

Figure 7: The geometric bit. (a) Two mirror states separated by a barrier of exactly one bond depth along the hinge path; the direct route through the repulsive core costs about thirty times as much, and a watershed over the full configuration space confirms no cheaper channel exists. (b) The same ε fixes retention through an Arrhenius law, so the memory's lifetime is set by the constant that binds the matter.

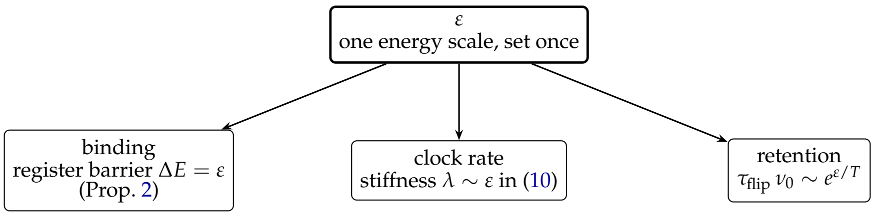

Figure 8: One constant, three bills. The same ε prices the matter's binding, the clock's rate, and the memory's lifetime, in fixed proportion — there is no separate constant for any of the three. This is a structural commitment rather than an economy of notation: any measurement that moved one of the three without the others would break the architecture, and Section 11 exhibits all three roles on one instrument.

is worth stating plainly, because it runs opposite to the usual framing of the measurement problem: in this model *collapse is the native operation and unitarity is the imported idealization*. What the substrate does for free is actualize; what it must borrow is reversible evolution between actualizations.

The bath is a real thermodynamic object. The stochastic element in the measurements of Section 8 is an Ornstein–Uhlenbeck thermostat on the field velocity, and it is not decorative: it equipartitions, delivering $\langle |v|^2 \rangle = 3T$ per site to within a few per cent, with an autocorrelation decaying at the declared rate. It is, however, *imposed*: the underlying dynamics is deterministic and supplies no ensemble of its own. Every probabilistic statement in this paper inherits that qualification.

Retention is Arrhenius, and this fixes the cost of memory. Proposition 2 put the barrier at one bond depth, and Section 5 gives $\tau_{\text{flip}} \sim \nu_0^{-1} e^{\varepsilon/T}$. A register is therefore not free: holding one bit for N clock cycles requires $\varepsilon/T \gtrsim \ln(N\,\nu_0 T_{\text{clock}})$, so memory depth is logarithmic in the energy budget — the familiar thermodynamic shape, arrived at here from the model's own statics rather than assumed. Figure 8 states the resulting economy as the falsifiable commitment it is.

And a timescale condition. The thermodynamic quantity that turns out to control whether actualization statistics are *quantum* is not an energy but a *ratio of timescales*: the mode frequency against the noise correlation time. It is the point at which the thermodynamic pillar stops being connective tissue and starts doing work.

The debt this leaves. Three purchases have now been made — a duration, a persistence, and a single constant ε that prices both — and with them a substrate can count. It can say that this happened, then that, and that the interval between them was so many ticks. What it still cannot say is *with what weight* one continuation occurs rather than another. Nothing in a clock, a barrier or an Arrhenius rate assigns a number to an alternative that did not happen.

That is a debt of a different kind from the previous three. A clock and a register were built; a weight cannot be built the same way, because the question is not what mechanism produces it but what *structure* the weights must have once frequencies can be accumulated at all — and accumulating frequencies is exactly what the last two sections bought. The next section answers it, and does so by importing rather than constructing. The import is itemized, because it is the largest single thing this paper does not derive.

7 Weights, and where they come from

7.1 States, effects, and convexity

Two preparations are equivalent if every subsequent experiment gives the same statistics; the equivalence class is a *state* ω. Two detector outcomes are equivalent if they have the same probability for every preparation; the class is an *effect* e. The primitive pairing is

$$e(\omega) \in [0,1], \tag{12}$$

read as the potential weight of actualizing e given ω. There is as yet no Hilbert space, no wavefunction and no Born rule. If ω_1 is prepared with frequency p and ω_2 with frequency $1-p$, operational consistency forces $e(p\omega_1 + (1-p)\omega_2) = p\, e(\omega_1) + (1-p)\, e(\omega_2)$, so the state space is convex and effects are affine. This is the general-probabilistic level: it contains classical probability, real-Hilbert quantum theory, and much else.

7.2 The reconstruction, imported

What selects complex quantum theory from that class is a separate, external result. Chiribella, D'Ariano and Perinotti [1] derive finite-dimensional complex quantum theory from the operational-probabilistic framework together with six principles: causality, perfect distinguishability, ideal compression, local distinguishability, pure conditioning, and purification — the last being what separates the quantum member from the classical one. Under those premises,

$$\mathcal{H}_A \simeq \mathbb{C}^{d_A}. \tag{13}$$

Status of (13). A theorem *external* to this paper, conditional on its six imported premises. Nothing here derives those premises from anything more primitive; Appendix B itemizes them as an import invoice, which is the honest replacement for treating "the Hilbert space" as a single unpriced assumption.

7.3 The trace rule from effect additivity

Once the quantum effect space is in hand, the shape of the weight functional is fixed by a second declared postulate.

Postulate 2 (Effect-noncontextual additivity)**.** A weight functional $w : \mathcal{E}(\mathcal{H}) \to [0,1]$ with $w(I) = 1$ satisfies $w(E+F) = w(E) + w(F)$ whenever $E + F \leq I$, for *all* effect decompositions, including those whose members do not commute.

Postulate 2 is substantive and load-bearing: it is a noncontextuality of valuation over generalized measurements, and a contextual valuation escapes what follows without contradiction. From additivity, $w(qE) = q\,w(E)$ for rational q. Monotonicity is derived rather than assumed — for effects $E \leq F$ the difference $F - E$ is an effect, so $w(F) = w(E) + w(F-E) \geq w(E)$ — and squeezing with rationals $q_n \uparrow \lambda \leq r_n \downarrow \lambda$ gives $w(\lambda E) = \lambda w(E)$ for real λ. Extending by differences yields a positive linear functional, and in finite dimension every such functional is a trace:

$$w(E) = \mathrm{Tr}(\rho_w E). \tag{14}$$

This is Busch's generalized Gleason theorem [2], valid already in dimension two, where the original Gleason theorem — which assumes additivity only over *projective* decompositions — requires dim ≥ 3 and fails. The strengthening from projectors to effects is exactly the content of Postulate 2. Throughout this paper, (14) and everything downstream of it is described as *derived from Postulate 2*, never as derived *simpliciter*.

7.4 Probability is state plus question

With (14) the logical order of Section 2 becomes formal. For a measurement $M = \{E_i\}$ with $\sum_i E_i = I$,

$$p(i \mid \rho, M) = \mathrm{Tr}(\rho E_i), \qquad \sum_i p_i = \mathrm{Tr}\,\rho = 1, \tag{15}$$

so a probability exists only once a question is supplied: the state is not itself a probability distribution. For a pure state and a sharp outcome this reduces to the familiar square,

$$p_i = \mathrm{Tr}\left(|\psi\rangle\langle\psi|\,|i\rangle\langle i|\right) = |\langle i|\psi\rangle|^2, \tag{16}$$

which in this architecture is a *theorem about weights* standing at the end of the chain $|\psi\rangle \to \rho \to (\rho, M) \to p_i$, rather than an axiom about probability standing at its beginning.

Two structural facts follow immediately and matter later. Global phase is unphysical, so pure potentiality is a ray and ψ decomposes into a weight structure and a phase structure, of which squaring retains only the first. And coherent potentiality is not ignorance over the same alternatives: for $|\psi\rangle = \alpha|A\rangle + \beta|B\rangle$ the cross term $2\,\mathrm{Re}\,\alpha^*\beta\langle A|E|B\rangle$ is absent from the corresponding mixture. That argument excludes one specific mixture reading; the stronger exclusion of ψ-epistemic models is the Pusey–Barrett–Rudolph theorem [4], at the price of its preparation-independence assumption.

7.5 Dynamics, instruments, and the selector

Requiring affineness on mixtures, positivity, and positivity under extension by an untouched ancilla gives complete positivity, so physical channels are CPTP maps in the instrument tradition of Davies and Lewis [3]. A reversible transformation preserving pure-state transition probabilities acts on rays and is unitary or antiunitary (Wigner [5]); for a one-parameter group the antiunitary branch is excluded because $U(t) = U(t/2)U(t/2)$ and the square of either is unitary, and the projective obstruction vanishes for $\mathbb{R}$ [6], so with strong continuity Stone's theorem [7] gives $U(t) = e^{-itG}$ and, on defining $H := \hbar G$,

$$i\hbar\,\partial_t|\psi\rangle = H|\psi\rangle, \qquad i\hbar\,\dot{\rho} = [H, \rho]. \tag{17}$$

The existence of a phase generator is structural; the numerical value of $\hbar$ is empirical and is not derived here. A measurement context is represented not by an observable but by an *instrument* $\{\mathcal{I}_o^M\}$ of CP trace-nonincreasing maps summing to a trace-preserving map, with $E_o^M = (\mathcal{I}_o^M)^*(I)$; an instrument carries two things at once, the weight of a possible actualization and the successor potentiality if it occurs.

What no part of this supplies is the selection of one outcome. Writing Σ_O for that block, there are two mathematically distinct implementations,

$$\underbrace{K_O(o \mid \rho, M, H)}_{\text{primitive stochastic}} \qquad \text{or} \qquad \underbrace{\lambda \sim \mu(\mathrm{d}\lambda \mid \rho, M, H), \;\; o = F_O(\rho, M, H, \lambda)}_{\text{context-complete deterministic}},$$

and in either case the physical content is the requirement that the resulting frequencies reproduce the weights,

$$(F_O)_*\mu(\{o\}) = \mathrm{Tr}(\rho E_o^M), \tag{18}$$

called the *Born pushforward* below. Note that Σ_O means the whole implementation — the kernel, or the pair (F_O, μ) — so that the measure cannot be hidden inside a deterministic map. Given an outcome, the successor state is ordinary conditionalization, $\rho' = \mathcal{I}_k^M(\rho)/p_k$; histories compose by $P(h) = \mathrm{Tr}[\mathcal{I}_{o_N}^{M_N} \circ \cdots \circ \mathcal{I}_{o_1}^{M_1}(\rho_0)]$, so the potential histories are plural while the actual history is singular. Collecting the pieces gives the recursion of Figure 1,

$$(H_n, \mathcal{C}_n, \mathcal{D}) \xrightarrow{\Gamma} \rho_n \longrightarrow \{(e_i, w_i)\} \xrightarrow{K \text{ or } (F,\mu)} e_{n+1} \longrightarrow H_{n+1}, \tag{19}$$

in which $\mathcal{C}_n$ is the preparation and control record and $\mathcal{D}$ the stipulated dynamics. The closure map Γ is part of the scaffold, not a theorem: history alone does not determine a state.

The weight layer is now complete, at a cost stated line by line. But nothing in it says *when* a weight becomes a frequency. That is not a question about the algebra of weights; it is a question about the mechanism that reads them out, and about how fast it reads compared with how fast the thing being read is moving. The next section measures exactly that ratio.

8 Born weighting as a regime

8.1 What a threshold counts

Consider the simplest possible actualization mechanism: a coherent field riding on noise, and an event recorded whenever the total crosses a threshold. Which feature of a mode does such a counter weight?

There are three candidates with different physical meanings. It might weight by *amplitude squared*, which is what a naive excess-rate calculation gives at leading order. It might weight by *energy*, which carries an extra factor of Ω^2. Or it might weight by *occupation* $n = E/\Omega$ — amplitude squared times frequency, the classical action variable and the adiabatic invariant — which is the Born-valued quantity, because the relativistic mode normalization $A \sim 1/\sqrt{2\Omega}$ is precisely the conversion between field amplitude and occupation. The physical intuition for the third is worth stating: a detection rate is presentations per unit time times success per presentation; success goes as amplitude squared and presentations go as frequency, and their product is occupation. On this reading the mysterious $1/\sqrt{2\Omega}$ of quantum field theory is the conversion from *how big the wave is* to *how often it knocks*.

Figure 9 draws the counting itself, deliberately without noise: two modes a threshold cannot tell apart by occupation — one tall and slow, one half the height knocking four times as often — and the two faces a noisy threshold multiplies, presentations counted at a low level and amplitude gating at a high one.

The three predictions are distinguishable. Preparing two standing modes at *equal occupation* and regressing the excess crossing rate on their spatial signatures yields a ratio R whose value is Ω_1/Ω_2 under amplitude weighting, 1 under occupation weighting, and Ω_2/Ω_1 under

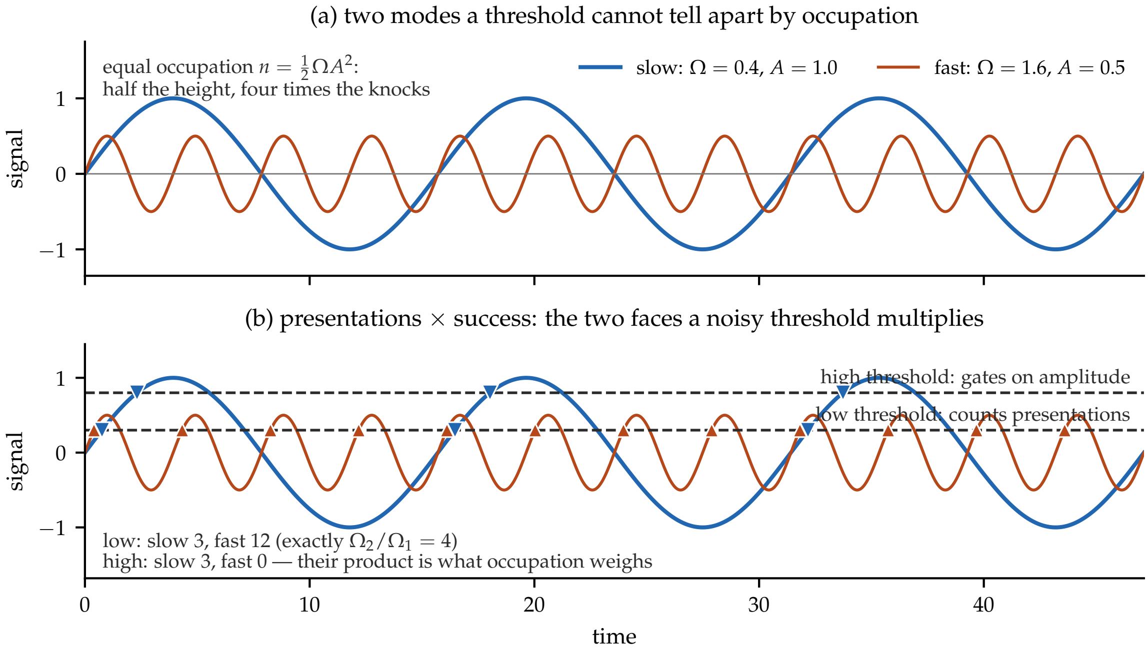

Figure 9: The mechanism, before any measurement. (a) Two modes at equal occupation $n = \frac{1}{2}\Omega A^2$: the fast one is half the height and presents four times as often. (b) The same traces against two thresholds, every crossing computed and marked. At the low threshold both modes clear, and the counts are exactly in the frequency ratio (12 against 3): crossings count *presentations*. At the high threshold only the tall mode reaches, and the fast mode logs zero: *success* gates on amplitude. A noisy threshold samples between these faces, and the product they multiply to — presentations × success — is occupation. This figure is an illustration of the counting logic on one declared trace, not a measurement; the measured interpolation between the two weightings is Figure 10.

energy weighting. We report the normalized statistic $\mathrm{BF} = (R - R_{\mathrm{amp}})/(1 - R_{\mathrm{amp}})$, the *Born-fraction*, which is 0 for amplitude and 1 for Born.

8.2 The measured interpolation

Under pre-registration — outcomes, gates, kill conditions and runner hashes locked before execution — the answer is neither of the three fixed alternatives. The weighting *interpolates*, and the control parameter is the product of mode frequency and noise correlation time. Across five arms spanning $\bar{\Omega}\tau$ from 0.64 to 78.36 the Born-fraction rises strictly monotonically from 0.049 to 0.836 (Table 1, Figure 10), with a descriptive crossover near $\bar{\Omega}\tau \approx 17$.
The reading, at exactly its scope: *within the declared mechanism class, Born weighting is the fast-mode/slow-noise asymptote of threshold statistics*. The rule appears as a regime, not an axiom. It is neither an exact Born law nor a theorem about any substrate; whether the asymptote is exactly 1 or near the descriptive 0.86 is open.

8.3 The negative, reported at equal prominence

The same law was then tested where it would matter more: on the model's own thermal field, with its own propagating noise, under a locked ten-cell design. **It did not transfer.** No declared axis reached the registered structure threshold — the Spearman correlation of Born-fraction against the primary axis was +0.01, against the secondary +0.37, and against threshold height −0.20, all far below the required 0.7. The registered outcome is negative.

Instrument	$\bar{\Omega}\tau$	Born-fraction	99% CI
saturation scan, arm 1	0.64	0.0494	[0.0359, 0.0623]
saturation scan, arm 2	2.54	0.1020	[0.0898, 0.1144]
saturation scan, arm 3	10.15	0.2735	[0.2549, 0.2903]
saturation scan, arm 4	19.59	0.4694	[0.4583, 0.4797]
saturation scan, arm 5	78.36	0.8362	[0.8130, 0.8612]
earlier discrimination, slow	≈ 0.7	0.024	[0.014, 0.036]
earlier discrimination, fast	≈ 2.7	0.099	[0.090, 0.109]
slow-channel test, $f = 1$	141.7	0.9361	[0.8953, 0.9700]

Table 1: Born-fraction against the frequency–correlation-time product. The two instruments are reported separately and are *not* averaged: the slow points differ by about 0.02 beyond seed noise, a declared phase-draw systematic. The last row is a pure slow channel rather than a sixth arm of the same scan.

The post-hoc diagnosis — and it is flagged as post-hoc, not registered — is visible in Figure 11: the mean-subtracted noise correlation time sat pinned between 0.6 and 1.9 ticks in every cell regardless of the thermostat's damping, so the accessible range of $\bar{\Omega}\tau$ was only 0.19–0.86, entirely below the crossover. Within that range the measured values, 0.01–0.05, are quantitatively consistent with the interpolation law. The structural suggestion is that thermal noise carried by the field itself can never be slow relative to that field's own band, because signal and noise are the same medium. We state that as a suggestion: the record disfavours the route in the sampled domain and does not prove that it can never succeed.

8.4 A slow channel, and the purity requirement

That diagnosis has a testable consequence: if the gate must be slow and the field cannot be, the model must be searched for a slow channel of different origin. It has one — a slowly accumulating potential sector sourced by the matter configuration. Measured under its own locked pre-registration on a self-gravitating configuration with no thermostat and no manifestation events, its fluctuations have correlation time $\tau = 317.9$ ticks against the field's 0.81: a ratio of 391, and $\bar{\Omega}\tau = 141.7$, deep in the regime the interpolation law identifies.

Feeding a channel at that timescale into the mechanism, with the fast field noise still present at full strength, gives the result in Figure 12: the Born-fraction stays pinned between 0.04 and 0.09 up to a slow-channel share of 75%, and reaches 0.936 only when the slow channel is essentially the *sole* stochastic element.

Proposition 3 (Purity requirement)**.** *Crossing statistics are dominated by the fastest stochastic component present. Born weighting therefore requires not merely the presence of a slow channel but its near-exclusivity at the threshold.*

This is a demanding condition and it is the honest headline of the whole exercise: a physical realization must *shield* the actualization comparison from the fast field, not merely add a slow term to it.

8.5 Three ceilings, and the gap nobody has closed

The architecture's central distinction has so far been qualitative: potentiality is *structured*, and actuality is singular. Bell correlations make the first half quantitative, and doing so exposes an open problem that is the paper's thesis in numerical form.

Three ceilings bound the CHSH quantity, and each corresponds to a different admissible explanatory structure:

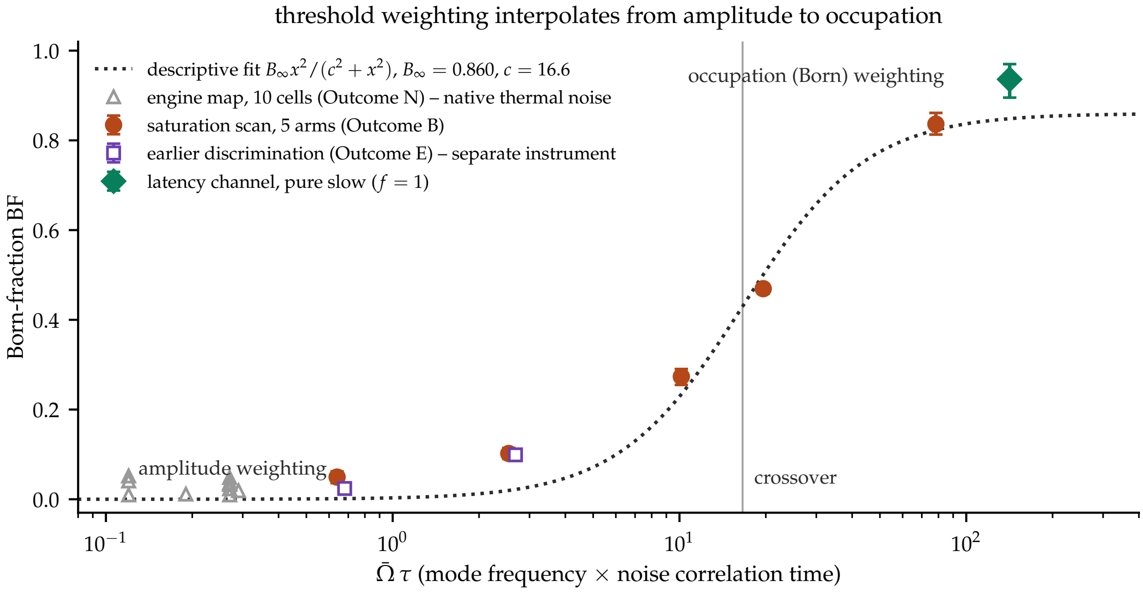

Figure 10: The central measurement. Threshold-crossing statistics interpolate from amplitude weighting to occupation (Born) weighting as mode frequency exceeds the noise bandwidth. Markers distinguish instruments deliberately: the two slow-point estimates are never averaged, the engine cells are a different noise structure spanning only $\bar{\Omega}\tau = 0.19$–0.86, and the highest point is a pure slow channel. The dotted curve is descriptive and carries no decision weight.

ceiling	value	structure that could produce it
local	2	an ensemble of pre-assigned values
quantum (Tsirelson [35])	$2\sqrt{2} \approx 2.8284$	a noncommutative weight structure
no-signalling (PR [36])	4	any consistent probability assignment

Three precisions keep the reading honest.

The first ceiling is not a bound on actuality. Every recorded outcome is actual whatever S turns out to be; actuality is singular per trial and is not bounded by anything. What 2 bounds is what *pre-assigned* values could produce — the sixteen vertices of the local polytope are exactly the deterministic assignments in which each system already carries an answer to both questions it might be asked. This is why singular actuality and $S = 2\sqrt{2}$ coexist without strain: the tension is with pre-assignment, not with actualization.

The second is a slice, not a shape. $2\sqrt{2}$ is the CHSH facet of the quantum set; that set is not a polytope and its full boundary is characterized only by a converging hierarchy of semidefinite relaxations [37].

The third is not merely a formal maximum. Correlations reaching 4 exist that never permit a signal — Popescu–Rohrlich boxes are perfectly consistent with relativistic causality [36]. So the outer ceiling is not an artefact of allowing the impossible; it is the honest extent of what consistency alone permits.

That last point is the substance. **Nothing in causality, locality, or logical consistency explains why nature stops at $2\sqrt{2}$.** Relativity licenses everything up to 4; the world uses 70.7% of it and leaves the rest unclaimed. Several principles have been proposed that recover Tsirelson's bound or approach it — information causality [38], macroscopic locality [39], local orthogonality [40], and limits on communication complexity [41] — and none is established as *the* derivation.

Figure 13 computes all three ceilings, and adds what the number buys. Device-independent certification turns S into a resource: at $S = 2$ the outcomes could have been written down

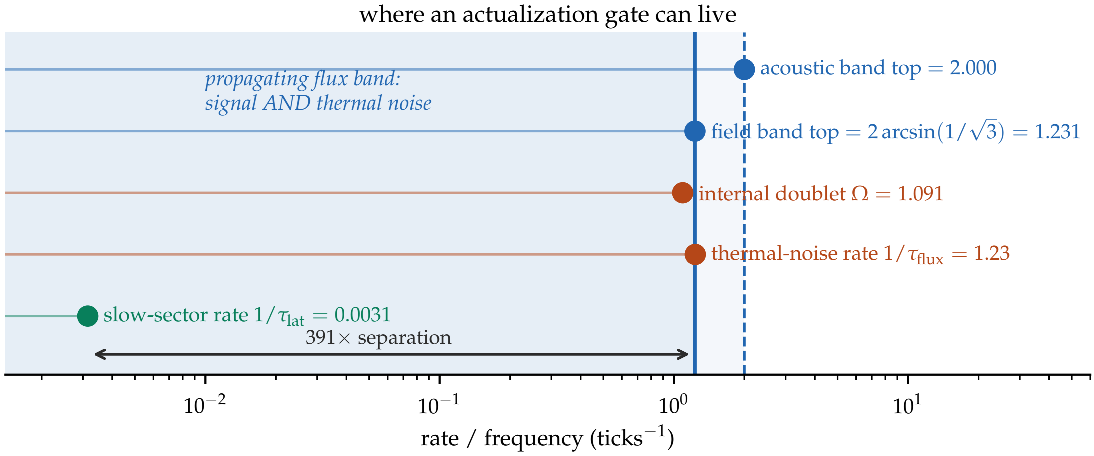

Figure 11: Why the negative was structural rather than accidental. The propagating band carries both the signal modes and the thermal noise, so the noise rate cannot be pushed below the band. Only a channel outside that band — here the model's slowly accumulating gravitational-potential sector, three orders of magnitude slower — sits where an actualization gate would have to sit.

in advance and the certified min-entropy is exactly zero, while at $2\sqrt{2}$ it is exactly one bit. Whatever fixes the ceiling therefore fixes how much unpredictability the world makes available — a physical quantity, not a bookkeeping one.

The architecture therefore acquires a quantitative form and an open problem in the same stroke. The potential is not "whatever is not forbidden": it is a specific, tightly constrained subset of the consistent, and the constraint has no accepted explanation. Whatever fixes the weight structure of Section 7 is the same thing that fixes this ceiling, and neither the reconstruction programme nor anything in this paper supplies it. The gap between $2\sqrt{2}$ and 4 is the width of that ignorance, measured.

8.6 What the angles do, and which part needs a record

One further separation is worth making explicit, because two claims are commonly run together and only one of them is true.

The CHSH number is a construction. Its four correlators are measured on four disjoint sub-ensembles; no trial contributes to more than one; and no individual context is anomalous — each sits at $1/\sqrt{2}$, comfortably inside the unit bound available to any correlator whatever. S is assembled, and it does not exist without something that assembles it. This is Fine's gluing statement in operational dress, and it is correct.

The noncommutativity is not a construction. It has consequences in single systems with no ensemble and no memory. Two crossed polarisers pass nothing; insert a third at 45° and $I_0/8$ passes. That is one beam making one pass, and the order of the elements changes what emerges. The same structure fixes ground-state energies — $E_0 = \hbar\omega/2$ follows from $[x, p] \neq 0$ and vanishes without it — which is why helium does not freeze at $T \to 0$ and why deuterium and hydrogen bonds differ in length and reaction rate. None of that requires a record.

The distinction matters for the architecture. The commutator is the *input* to $S^2 = 4 + \|[A_0, A_1]\|\|[B_0, B_1]\|$, and S is the *readout* (Figure 14). Records are required for the readout and not for the commutator, and the ceiling of $2\sqrt{2}$ was fixed by the algebra before any apparatus existed to approach it. An account that locates noncommutativity in the compiled record therefore explains the wrong object: it would leave the three-polariser experiment, and the stability of ordinary matter, without a cause.

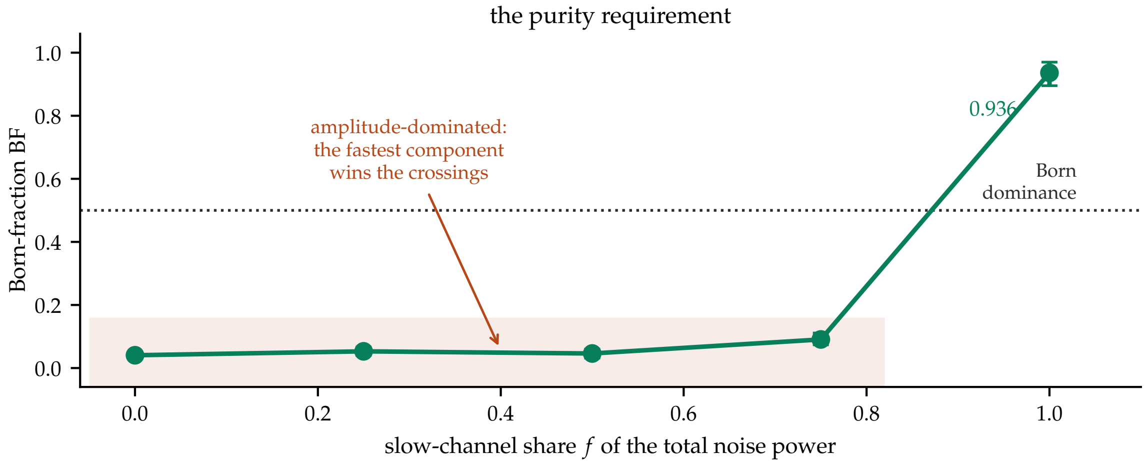

Figure 12: The purity requirement. Mixing a slow channel into fast noise does not gradually produce Born weighting; the statistics remain amplitude-dominated until the slow channel is nearly exclusive.

Scope of Section 8. All ensembles are imposed; the mechanism-level measurements run on a standalone instrument rather than the full model; the coupling in the last experiment is a declared additive model class. The slow-channel measurement's profile contains no random-number consumer, so its run is fully deterministic and its seed-independence is vacuous — configuration-varied robustness is a registered refinement, not a completed check. None of this establishes a Born law for any substrate. The tier structure of Section 8.5 is a restatement of established bounds, not a new result; its contribution is to locate the paper's open question inside them.

Everything established to this point is a *one-frame* statement: a regime condition read on one instrument, a ceiling fixed by one algebra, a clock ticking where it was built. Relativity is the demand that two frames agree, and it is the first obligation in this paper that cannot be discharged by any single apparatus, however good.

9 Relativity

The architecture divides cleanly under relativity, and the division is the paper's organizing claim about what merges and what does not.

9.1 The weight layer relativizes; this is standard physics

Covariant potentiality is the assignment of local algebras to spacetime regions with Einstein causality, $[\mathcal{A}(O_A), \mathcal{A}(O_B)] = 0$ for spacelike-separated regions [13, 14], with probabilities given by the state–effect pairing $p_\omega(E) = \omega(E)$ — the representation-independent replacement for the trace, since a generic local algebra is not of type I and admits no intrinsic density matrix. Covariant evolution of the potential layer is the Tomonaga–Schwinger equation [16], conditional on integrability. And for causally disjoint local instruments the joint statistics are independent of composition order,

$$p_\omega^{A \prec B}(a, b|x, y) = p_\omega^{B \prec A}(a, b|x, y), \tag{20}$$

provided the instruments satisfy a causal local measurement scheme [15]. Equation (20), not trace preservation alone, is the operational statement that the order of spacelike actualizations is gauge at the level of ensembles.

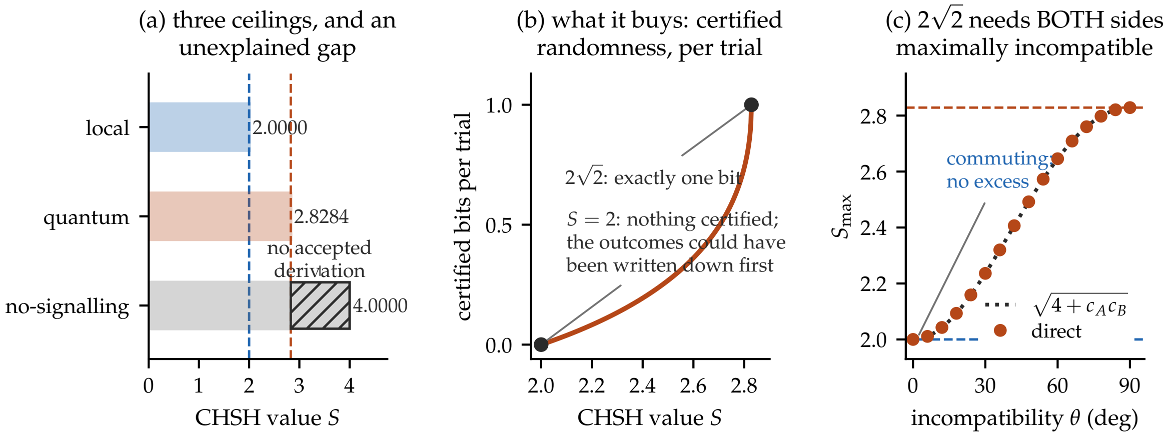

Figure 13: The three ceilings, computed. (a) Local correlations stop at 2, quantum at $2\sqrt{2}$, and no-signalling at 4; the hatched band is the 29.3% that relativistic consistency permits and nature does not use. (b) What the number buys: the device-independent min-entropy runs from exactly zero certified bits at $S = 2$ — where the outcomes could have been written down in advance — to exactly one bit at $2\sqrt{2}$. (c) The price, from $S^2 = 4 + \|[A_0, A_1]\|\,\|[B_0, B_1]\|$: each commutator norm is capped at 2, so a product of 4 requires both at maximum. There is no partial-credit route to $2\sqrt{2}$. The dotted curve is the identity and the markers are the directly computed operator norm; they agree to 2×10^{-15}.

Order-independence is not selector covariance. It is tempting, and wrong, to conclude that a covariant weight layer delivers a covariant selector. Let μ_f be the measure over complete realized histories induced by a selector law under foliation f, and let $\tau_{f\to g}$ exchange only the birth order of spacelike-incomparable events. The two genuinely distinct further conditions are

$$\mu_g = (\tau_{f\to g})_* \mu_f \qquad \text{and} \qquad (\pi_{\mathcal{E}})_* \mu_{f_0} = P^{\mathrm{cov}}_{\mathcal{E}} \ \text{ for every experiment } \mathcal{E}, \tag{21}$$

the first requiring genuine equivariance and the second requiring only that a physically fixed preferred foliation f_0 be operationally hidden. **Neither follows from** (20), which contains no selector law and no history measure. Keeping these apart is, in our view, the single most common conflation in discussions of relativistic collapse.

9.2 Four representative responses, each priced

The available responses are families rather than an exhaustive classification, and they are not mutually exclusive.

No selector. Deny Postulate 1: unitary theory only, and the covariance problem disappears because there is nothing to make covariant [8]. The price is symmetric and real — the Born weights must then be derived without a selector, whether by decision-theoretic argument [9, 10] or envariance [11], and both remain contested [12].

Hypersurface-relative conditionalization. Take the state update to be conditionalization and let it be hypersurface-relative, as in [17, 18]. This is update semantics; it does not by itself say which event becomes actual or with what measure.

Covariant stochastic flashes. Take the actualization events themselves as the primitive ontology [19], for which relativistic models exist for fixed particle number, originally without interaction [20] and subsequently with interaction [21]. Indistinguishable particles, variable number, and full interacting field theory remain open.

A real preferred foliation. Let the selector act on a real but hidden foliation. This makes a total actualization order well defined, and it is where the discrete model of this paper sits, since a global tick order is one of its axioms. The price is usually stated as two independent

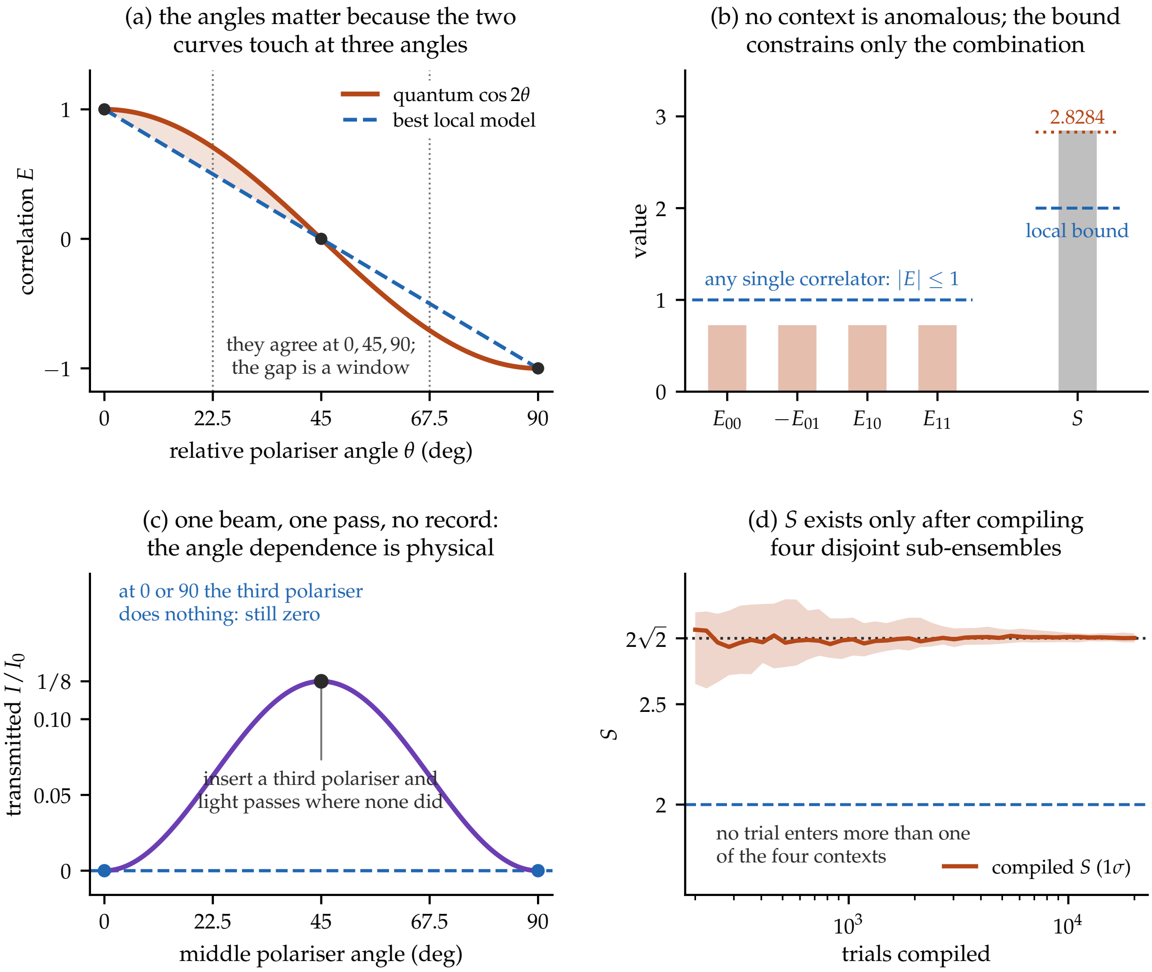

Figure 14: The angles, and the boundary. (a) For polarisation-entangled light the quantum correlation $\cos 2\theta$ and the best local model — a straight line — *touch* at 0°, 45° and 90°, so the violation is an angle window rather than a uniform excess; the widest single gap is 0.211 at 19.85°. (b) At the CHSH settings each of the four contexts contributes exactly $1/\sqrt{2}$, well inside the $|E| \leq 1$ available to any correlator: no context is individually anomalous, and the local bound constrains only the total. (c) Two crossed polarisers transmit nothing; inserting a third at 45° transmits $I_0/8$. One beam, one pass, no ensemble comparison and no record — the angle dependence is physical, not a feature of the bookkeeping. (d) S by contrast is a compiled statistic: the four sub-ensembles are verified disjoint and exhaustive, no trial enters more than one, and the number exists only after all four are combined.

proofs: that the selector law satisfies (21), and that every observable sector passes ordinary Lorentz-recovery tests. Section 9.3 shows that these two are not on the same footing — the first dissolves, and the second half-closes.

9.3 Discharging the first debt, and locating the second

The two proofs owed by a preferred-foliation ontology are not symmetric.

Proposition 4 (Operational hiding is not an independent obligation)**.** *Let a selector on a fixed foliation f_0 generate a measure μ_{f_0} over complete histories. If* (i) *the marginal of μ_{f_0} on recorded data equals the quantum composed-history weight $P(h) = \mathrm{Tr}[\mathcal{I}_{o_N}^{M_N} \circ \cdots \circ \mathcal{I}_{o_1}^{M_1}(\rho_0)]$ for every experiment, and* (ii) *those weights satisfy* (20)*, then $(\pi_{\mathcal{E}})_*\mu_{f_0} = P_{\mathcal{E}}^{\mathrm{cov}}$ for every operational experiment: the foliation is operationally hidden.*

Proof. By (i) the recorded distribution is $P(h)$. By (ii) $P(h)$ is unchanged when spacelike-separated instruments are exchanged in composition order, and two admissible foliations of one experiment differ exactly by such exchanges; so $P(h)$ is foliation-independent and equals $P_{\mathcal{E}}^{\mathrm{cov}}$. □

The proof is nearly immediate once the hypotheses are stated, and its value is relocation rather than resolution: hiding a preferred foliation requires nothing beyond reproducing quantum statistics. It is one-way — foliation-independent statistics need not be quantum — and it does not deliver the stronger equivariance of (21)'s first equation, which concerns unobservable history detail and is presumably false for a genuinely preferred foliation. Neither is needed for empirical adequacy.

The corollary is more informative than the proposition.

Proposition 5 (The barrier is not relativistic)**.** *If a substrate's four CHSH observables are functions on one sample space under a single setting-independent measure, their pushforward is a joint distribution and Fine's theorem gives $|S| \le 2$. Hypothesis* (i) *of Proposition 4 then fails on Bell experiments, and with it operational hiding.*

Exhaustive enumeration gives $\max |S| = 2$ over all sixteen deterministic joint assignments against the attained quantum $2\sqrt{2} = 2.828427$, while the order-independence half of Proposition 4 checks numerically to 2.2×10^{-16} over random states and settings. So: *if a substrate reproduced quantum statistics, its preferred foliation would automatically be hidden.* The foliation is not the difficulty. Reproducing the statistics is, and that obstacle is a Bell-theoretic one which has nothing to do with relativity — it would be present in a non-relativistic world. Figure 15 draws the two propositions as one diagram: what buys hiding, and where the one blocked path actually is.

The second debt is real, and its free sector can be paid exactly. The model's Laplacian is the eighteen-point operator $\frac{1}{3}\sum_{\text{face}} + \frac{1}{6}\sum_{\text{edge}} - 4$, whose symbol expands (in exact arithmetic) to

$$L(\mathbf{k}) = -k^2 + \tfrac{1}{12}(k^2)^2 - \tfrac{1}{360}\Big(\textstyle\sum_i k_i^6 + 5\sum_{i\neq j} k_i^4 k_j^2\Big) + O(k^8). \tag{22}$$

The fourth-order term is a perfect function of k^2: **the stencil is isotropic through fourth order**, and structurally so — the naive seven-point operator has fourth-order coefficient $\frac{1}{12}\sum_i k_i^4$, which is not, and the twelve edge terms are exactly what cancels it. The leading anisotropy is therefore pushed to sixth order, giving a fractional direction-dependent phase velocity

$$\left|\frac{\Delta v}{v}\right|_{\langle 100\rangle - \langle 111\rangle} = \frac{(ka)^4}{3240} + O\big((ka)^6\big). \tag{23}$$

At Planck spacing this is 1.4×10^{-56} for the highest observed gamma rays and 1.4×10^{-36} for the most energetic cosmic-ray primaries, tens of orders of magnitude below optical-cavity isotropy bounds near 10^{-18}.

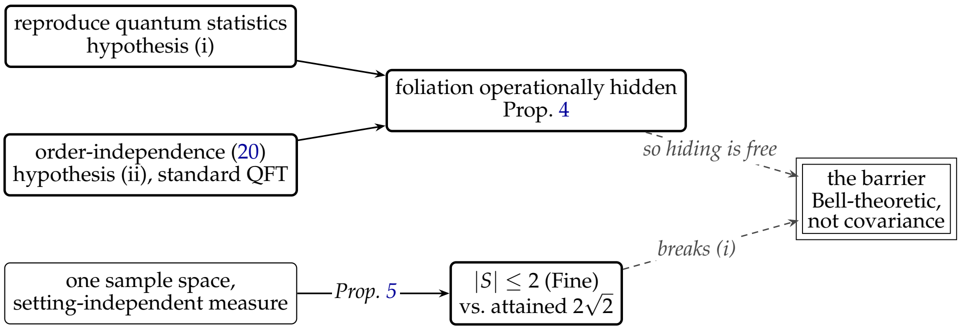

Figure 15: The two reductions in one picture. Hiding a preferred foliation follows from reproducing quantum statistics together with standard order-independence — it is not an independent obligation. What blocks a classical substrate is the lower path: a single setting-independent sample space caps CHSH at 2 against the attained $2\sqrt{2}$, which breaks hypothesis (i) directly. The open obligation (double-ruled) is therefore Bell-theoretic and would exist in a non-relativistic world.

Radiative stability, reduced. Equation (23) is a tree-level statement, and a small tree-level number is not by itself evidence of Lorentz recovery. The standard objection is that a Planck-suppressed dimension-six violation, inserted in a loop whose integral is dominated by the cutoff, generates a *dimension-four* violation with coefficient of order α/π, requiring tuning of order 10^{-20} against observation [22]. Three exact facts change what that argument demands of a theory built on a single substrate.

First, the isotropy is forced. For a general cubic-symmetric eighteen-point operator with face weight w_f and edge weight w_e, the continuum limit requires $w_f + 4w_e = 1$ and isotropy of the k^4 term requires $w_f = 2w_e$; together these have the unique solution $w_f = \frac{1}{3}$, $w_e = \frac{1}{6}$ — the production weights. There is no free parameter to be tuned.

Second, the dimension-four anisotropy is forbidden, not small. Averaging a general symmetric quadratic form $c_{ij}k_ik_j$ over the forty-eight elements of the cubic group leaves $c_{ij} \propto \delta_{ij}$. Any cubic-invariant quadratic form *is* isotropic, so the dangerous coefficient vanishes identically.

Third, and decisively, dimension-four violation is relational. For a single propagating sector with $\omega^2 = c_0^2(1+\epsilon)k^2$, the rescaling $t \to t/\sqrt{1+\epsilon}$ removes ϵ completely: that sector's speed *is* the definition of the speed, and there is nothing to compare it against. With two sectors the same rescaling leaves $(1+\epsilon_2)/(1+\epsilon_1)$, so only the difference survives. This is why every experimental bound on dimension-four violation is a bound on a relative quantity — photon against matter, or species against species.

Proposition 6 (Reduction of radiative stability)**.** *The radiative-stability problem concerns* differential *renormalization: loops in one sector shifting its cone relative to another's. Its observable content therefore requires at least two independently renormalized propagating sectors. A theory whose excitations are all configurations of one substrate field inherits instead the question of whether its emergent sectors share a cone.*

Nor is the surviving dimension-six term enhanced: the Collins mechanism gains a factor Λ^2 by converting a $1/\Lambda^2$ coefficient into a dimensionless one, and a dimension-six operator renormalizing itself gains nothing, so corrections stay of order $(\alpha/\pi)a^2$ — the same order as (22)'s tree value.

What remains, stated as a deferral rather than a solution. Proposition 6 does not solve the naturalness problem for multi-species effective theories; it identifies which part of it a single-substrate theory actually carries. That part is the common-cone question: the matter and field sectors of the discrete model propagate by different mechanisms, no interacting common cone is established, and the wave speed $1/\sqrt{3}$ is a selection rather than a forced saturation. Section 9.5.2 settles the *free* half of it constructively and states the price; interacting vertices and operational composite boosts remain untested. And the deferral has a date: the moment a second independently renormalized sector emerges — charged matter with its own radiative structure — the problem returns in full, and nothing argued here will avert it. Physical Lorentz invariance remains open.

9.4 Where the remaining obligation lives: one body or two

The two reductions above have the same shape, and it is worth asking why. Both dissolved an apparently relativistic obligation into something that was not about relativity: hiding a foliation reduced to reproducing quantum statistics, whose barrier is Bell-theoretic; radiative stability reduced to a comparison between sectors. The common cause is a structural feature of the Lorentz group itself.

Rotations are the little group of a timelike vector: acting on a body's rest four-velocity $u = (c, \mathbf{0})$, a spatial rotation leaves it fixed. A boost does not — it maps that rest frame to a different one. So the group's two factors differ in what it takes to test them. A rotation can be tested by *one* apparatus, reoriented and compared against itself. A boost relates two rest frames, and a rest frame must be physically instantiated by a clock and a rod; testing boost invariance therefore requires a second body. The decomposition is exhaustive and has no overlap:

	one body (rotations)	**two bodies** (boosts)
what is compared	apparatus with itself, reoriented	one frame against another
typical test	isotropy, sidereal variation	time dilation, cone equality
status here	**closed**	**open**

This is exactly the seam along which our results fell. The argument that closed the dimension-four anisotropy was a statement about the point group O_h — a rotation group — and the surviving residual (23) is a one-body observable: reorient the apparatus and look for a sidereal signal. Everything still open is a boost statement, and each is two-body: whether the matter and field cones agree, whether interacting vertices preserve a common cone, whether composite clocks and rods transform correctly. The single-cone absorption argument of Section 9.3 says the same thing from the other side: one cone can always be scaled to unity, while a *ratio* of two cones is invariant under every global rescaling and cannot be defined away.

Proposition 7 (The boost half is clock-gated)**.** *Every boost observable is a functional of at least two proper-time parametrizations — time dilation compares* $\mathrm{d}\tau_1$ *with* $\mathrm{d}\tau_2$*; velocity composition relates two frames; geodesic deviation is defined by the separation of two worldlines, a single freely falling body detecting nothing by the equivalence principle. A theory that has not constructed a physical clock therefore cannot pose the boost tests at all: their observables are undefined in it, not merely unverified.*

Proposition 7 reorganizes the recovery contract rather than discharging it. The one-body register is closed, with numbers. The two-body register is open, and it is gated on precisely the construction Section 4 priced and the native search did not find. This is the second time the same object has gated something: Section 3 argued that if the stage index is to mean elapsed time, something physical must carry it, and Section 10 will pay that argument out in full. The same instrument is the precondition of the boost question. One construction, two obligations, both waiting on it — which is a more useful statement of where this programme is stuck than "Lorentz invariance is open," because it names the object whose absence is the obstruction.

The reading extends past special relativity. Curvature is defined operationally by geodesic deviation, so gravity's observable content is two-body as well, and the same gate applies to it.

One component of the two-body register can be sharpened immediately. The model's wave speed $1/\sqrt{3}$ is not the stability bound: for the eighteen-point operator the symbol attains $\max|L| = 16/3$ on the face diagonal, giving $C \leq \sqrt{3}/2$. It is instead exactly the largest isotropic speed whose light cone is *contained* in the octahedral causal polytope, whose facets $\sum_i \pm x_i = 1$ lie at distance $1/\sqrt{D}$ from the origin. That is the same $1/\sqrt{D}$ as the per-axis component of an isotropic unit spread, since $\mathbb{E}[n_i^2] = 1/D$ for a uniformly random unit vector — one isotropic unit distributed over D orthogonal axes, counted two ways. The characterization is exact, and it relocates the choice rather than removing it: the containment bound is $1/\sqrt{3}$ only for six-neighbour causality, whereas the eighteen-point Laplacian's reach and the model's stated Moore locality both give 1. The cone speed is therefore an interior selection whose adoption carries a commitment about which neighbourhood defines causal reach.

That commitment turns out to be lighter than it looks, and it is worth saying exactly how light, because the temptation is to record it as an open empirical question. It is not one. The competing neighbourhoods differ only in where the field's *exact zero* begins, and the field never approaches that boundary: measured on the point-source Green's function, the amplitude at the strict support edge falls by 1.19 decades per tick — 2.5×10^{-7} of the peak at $t = 8$, 2.7×10^{-17} at $t = 16$, 2.0×10^{-26} at $t = 24$ — crossing double-precision representability at about sixteen ticks. With the operator held fixed, no measurement and no simulation can distinguish six-neighbour from eighteen- or twenty-six-neighbour causality. The choice is definitional, about a boundary of measure zero in any observable, and should not be presented as awaiting evidence.

What *is* observable is set by the dispersion rather than by the stencil, and it is reassuringly isotropic. The strict support is a cuboctahedron, anisotropic by 34.3% between its face and edge directions; the field's own causal surface is a sphere to 0.71% at the signal front, and even fifteen decades down in amplitude reaches only 7.2% — so the stencil's shape is never attained at any measurable amplitude. The residual anisotropy also *decreases* with elapsed time, as $t^{-1.1}$ to $t^{-1.3}$ depending on probe depth, consistent with the Airy-front expectation $t^{-4/3}$ for an $O(k^4)$ dispersive correction. Isotropy is restored in the continuum limit at a measured rate.

None of this closes anything. It is a one-body statement — an independent cross-check of the free-sector result above by a different method — and the obligation that matters is the two-body one.

9.5 The two-body bill, itemized

Six line-items follow. Three are computed and close at their stated scope — the free massive modes, the composite inheritance, and the one-energy dilation; one is resolved in principle and priced but not built — the cross-sector cone; and two are diagnoses that relocate the remaining work — what a potential is, and what the substitution shows. Each names the object whose absence blocks what is still open.

9.5.1 Free massive modes

That obligation, however, was bundled too coarsely, and unbundling it turns part of it into a calculation.

Begin with what will *not* work. The natural test is to take the clock of Section 4, give it momentum, and compare its period to γ. It is vacuous, for the reason recorded when that clock was bought: a mechanical framework is Galilean invariant and reports zero dilation whatever the substrate does. Dilation in a field substrate comes from the binding being mediated at the substrate's own speed — Lorentz's electron-theory reasoning, and the light-clock argument in modern dress — so the object to interrogate is the *dispersion of a massive mode*, which needs no clock at all.

The operational form is exact. A packet around wavenumber k moves at its group velocity $v_g = |\nabla_k \Omega|$, and the relativistic prediction for its internal rate per unit proper time is

$$\Omega_{\text{proper}}(k) = \Omega(k)\sqrt{1 - v_g(k)^2/c^2} = \text{const}, \tag{24}$$

an identity whenever $\Omega^2 = c^2k^2 + M^2$. So (24) is k-independent precisely when a moving clock of that species dilates exactly, and its k-dependence *is* the violation. Adding a mass to the leapfrog gives the exact dispersion $4\sin^2(\Omega/2) = C^2(-L(k)) + M^2$, and expanding along [100], [110] and [111] returns *identical* coefficients through k^4:

$$\Omega^2 = \left(M^2 + \tfrac{M^4}{12}\right) + \left(\tfrac{1}{3} + \tfrac{M^2}{18} + \tfrac{M^4}{90}\right)k^2 - \left(\tfrac{1}{54} + \tfrac{M^2}{1080}\right)k^4 + \dots$$

Reading the k^2 coefficient as the squared limiting speed,

$$\frac{C_{\text{eff}}(M)}{C} = 1 + \frac{M^2}{12} + \frac{19M^4}{1440} + O(M^6). \tag{25}$$

The limiting speed is thus *exactly isotropic* — the three symmetry directions agree identically, not approximately — and *depends on the rest mass*. The second is a negative result: two species of different mass do not share a cone exactly. It is the classic species-dependent maximum-attainable-velocity signature.

Within a species the situation is benign. Using that species' own C_{eff}, the residual in (24) is $O(k^4)$ and equals $k^4/(36M^2)$ to three or four significant figures at every mass sampled; in the physically natural variable $\beta = v_g/C$ this is

$$\frac{\Delta\Omega_{\text{proper}}}{\Omega_{\text{proper}}} \simeq \frac{M^2\beta^4}{4}. \tag{26}$$

Each mass therefore carries its own very nearly Lorentzian sector, and the violation lives in the mismatch *between* sectors. Numerically, with one tick at 3.11×10^{-44} s, the electron and proton limiting speeds differ by 1.6×10^{-40}, and an electron clock at $\beta = 0.1$ dilates correctly to 1.5×10^{-50}. This is a dimension-six-suppressed, $(m/M_{\text{Planck}})^2$ effect — a concrete instance of the "returns at dimension six" noted in Section 9.3, here at tree level.

The same expansion reproduces the free-sector anisotropy of Section 9.3 independently, giving $(v_{[100]} - v_{[111]})/C = -k^4/3240$ exactly, which validates the chain.

One scope limit matters more than the result: this concerns *one sector with a varying mass*, whereas the far larger problem is that distinct sectors of the model carry leading speeds differing at order unity. That problem is the subject of the next subsection, and it turns out to be the one place in this paper where a stubbornly negative result inverted.

9.5.2 A common cone across sectors

One sector can always be given unit speed by choosing the time unit. That removes one normalization and no more: if two propagating sectors have slopes c_A and c_B, the ratio c_A/c_B survives every common unit change and is physical preferred-frame data. The discrete model carries $c^2 = 1/3$ in its production field sector and raw 1 in a standalone Wilson matter sector — an order-unity mismatch, against which the 10^{-40} of the previous subsection is irrelevant. This is the real obligation.

The first three findings are negative, and sharply so. Writing $u_i = \cos q_i$, matching a scalar Wilson parameter r to the field symbol identically, $c_s^2[3 - \sum u_i^2 + r^2(3 - \sum u_i)^2] = \kappa\,(-L_{18})$, gives ten coefficient equations of which two suffice: the u_1u_2 coefficient forces $\kappa = -3c_s^2r^2$ and the u_1 coefficient forces $\kappa = +9c_s^2r^2$, so $c_s^2r^2 = 0$. *No real* (c_s, r) *matches at any order.* Second, the O_h-invariant polynomials are generated by $\sum q_i^2$, $\sum_{i<j} q_i^2q_j^2$ and $q_1^2q_2^2q_3^2$, so the number of isotropy conditions grows without bound — cumulatively $0, 1, 3, 6, 10, 16, 23, 32$ through order q^{16} — and matching is a Symanzik-style improvement programme with an unbounded bill rather than a one-off purchase. Third, and least expected, enriching both structures with a face-diagonal shell yields six equations in seven Gram monomials, which *solve*; what fails is realizability. For a real kinetic vector the Gram $A_{st} = a_sa_t$ is an outer product and must have $A_{11}A_{22} - A_{12}^2 = 0$, whereas the solution forces $A_{11}A_{22} - A_{12}^2 = \frac{2}{9}\kappa^2$. The required Gram has rank two; the obstruction is structural, not arithmetic, and being proportional to κ^2 it survives every rescaling. Adding the fifth Clifford structure available to four-component spinors leaves the residual unchanged; adding a third kinetic shell makes the rank conditions insoluble.

What dissolves all three at once is a change of lattice. Regrouping and applying $1 - \cos A = 2\sin^2\frac{A}{2}$ and $1 - \cos A\cos B = \sin^2\frac{A-B}{2} + \sin^2\frac{A+B}{2}$ exhibits the field symbol as a *sum of squares*, and every argument that appears is a half-angle. Squares are hopping operators and their arguments are displacements, so the field symbol is the square of an operator on the lattice of *half* the spacing. Wilson fermions hop by integer displacements and no amount of shell-adding moves them off that sublattice. The sectors were being matched on the wrong lattice — which is the reason behind the exact no-go and the rank obstruction rather than a restatement of them.

The price is then a counting question, since $H = \sum_\mu \Gamma^\mu\phi_\mu$ squares to $\sum_\mu \phi_\mu^2$ only when the Γ^μ mutually anticommute, and dimension 2^k carries $2k+1$ such structures. Here the first decomposition we found was not the shortest, and the difference matters by a factor of four. Restricting to nearest-neighbour hops on the half-offset lattice — degree at most one per variable in $x = q/2$ — and writing $s_i = \sin x_i$, $c_i = \cos x_i$, there is an exact and cubic-covariant decomposition into *seven*:

$$-L_{18} \;=\; 4\sum_i s_i^2c_j^2c_k^2 \;+\; \tfrac{16}{3}\sum_i s_j^2s_k^2c_i^2 \;+\; 4\,s_1^2s_2^2s_3^2, \tag{27}$$

with $\{i,j,k\} = \{1,2,3\}$. It is checkable by hand: with $A = s_1^2$ etc., $S_1 = A + B + C$ and $S_2 = AB + BC + CA$, the three groups contribute $4[S_1 - 2S_2 + 3ABC]$, $\frac{16}{3}[S_2 - 3ABC]$ and $4ABC$; the triple products cancel exactly, $12 - 16 + 4 = 0$, leaving $4S_1 - \frac{8}{3}S_2 = \frac{4}{3}[3S_1 - 2S_2] = -L_{18}$. The lone singlet is not decoration — it is what kills ABC. A search over the same class finds decompositions at every length down to *four*, certified to 5.3×10^{-15} on four thousand fresh momenta, and none at three — where three is also the rigorous floor, since the symbol's Hessian at a zero has rank three and n squares can produce at most rank n. The four is genuinely non-covariant, so four and seven are the honest bracket. Seven costs spinor dimension eight, four costs four. (An earlier draft of this passage reported the minimum as five; its search drew four hundred starts from a single initialization scale, which samples one basin many times rather than many basins once. Dispersing the scale finds the four within tens of starts. The correction is retained here because the method failure is as instructive as the number.) Figure 16 draws the mismatch, its cause, one decomposition, and the price.

Which half-offsets appear is the question that decides what this costs the primitive ontology, because displacements are positions. Three shells are available — face $(\pm\frac{1}{2}, 0, 0)$, edge $(\pm\frac{1}{2}, \pm\frac{1}{2}, 0)$ and body centre $(\pm\frac{1}{2}, \pm\frac{1}{2}, \pm\frac{1}{2})$ — and the restriction now cuts finely: confined to body centres alone the search reaches five and stalls there over two thousand dispersed starts, while the unrestricted four uses the face and edge shells. So the shortest carrier buys one square by leaving the body centres, and the body-centre-only reading costs one square of length. Either way the carrier does not want a finer cubic lattice — every shell in play is a half-offset of the existing one — and the lattice spacing, on which every dimensional prediction depends, does not move. (The stall at four is search evidence of the same kind that previously mis-reported the minimum, and is offered with exactly that confidence; the rank bound $n \geq 3$ is the only proof in this paragraph.)

Seven is moreover not a search outcome but a dimension. The eight half-angle triple products $\prod_i g_i(q_i/2)$ with $g_i \in \{\cos, \sin\}$ satisfy $\sum(\prod_i g_i)^2 = \prod_i(\cos^2 + \sin^2) = 1$ identically. One of the eight is the pure cosine $S = \prod_i \cos(q_i/2)$ — which is exactly the body-centred-cubic nearest-neighbour structure factor, the eight neighbours sitting at $(\pm\frac{1}{2}, \pm\frac{1}{2}, \pm\frac{1}{2})$ — and the remaining seven *are* the carrier sector, so their equal-weight sum is $1 - S^2 = (1 - S)(1 + S)$. Equation (27) is the diagonal decomposition of that sector, one square per basis element, which is why it is covariant and why it is exactly seven. The field symbol and the body-centred symbol are two points in one seven-dimensional cone, differing only in their weights.

Two consequences deserve stating, one structural and one from Clifford counting. Structurally, this is the third independent route by which the discrete model arrives at the body-centred lattice, the other two being a Watson-type period identity and the emergence of an SU(3) factor, both from the same triple cosine product at the same half arguments; and it correspondingly *sharpens* a known tension, since the model's production stencil is orthogonal to that lattice. The counting: a mass term must anticommute with every kinetic structure, so it costs one further structure, and saturation alternates down the ladder — four squares at dimension four leave *one structure spare*, so the minimal carrier *admits a mass at ordinary Dirac dimension*, while the covariant seven saturates dimension eight exactly, so cubic covariance together with a mass forces sixteen. Covariance, not mass, is what costs spinor dimension here. (An earlier draft, built on the mis-reported five, concluded the opposite — that the minimal carrier was necessarily massless; that prediction died with the five and is recorded here so it cannot be cited from memory.)

> **What this does and does not close.** It closes the question of whether a common cone is obstructed in principle: it is not, there is an explicit exact decomposition, and its price is stated in physical currency. It does not deliver a working matter sector. Whether the resulting operator is an acceptable fermion — doubling, chirality, and the locality of the induced interactions — is untested; the time discretization is a separate matching problem that can only add conditions; a spinor's worth of anticommuting structures is not a ternary state, so the carrier must either *emerge* from the primitive states or be *adopted* as a new type at a price; and the order-unity mismatch still stands in the model as it is actually run. Interacting vertices remain open, as Proposition 7 says they must.

9.5.3 What a composite inherits

The remaining question — whether a bound composite inherits its constituents' limiting speed — is ill-posed until one says what makes it a composite. A body is bound by something, and the binding decides which constituents' properties are averaged and with what weights. That vague statement has an exact form. Treating $\delta_a = (c_a^2 - C^2)/C^2$ as a perturbation on a Lorentzian theory, the composite's shift is $\langle \Delta H \rangle$ in the bound state, and the K-dependent part of each constituent's kinetic energy is its share of the total momentum:

$$\delta_{\text{comp}} = \sum_a w_a \delta_a, \qquad w_a = \frac{\mu_a}{\sum_b \mu_b}, \quad \mu_a = \frac{m_a}{c_a^2}. \tag{28}$$

The weights need no interaction model: a composite at total momentum K distributes it by the co-moving condition, which is exactly $E(K) = \min\{\sum_a \omega_a(k_a) : \sum_a k_a = K\}$, and to first order in δ the binding shifts the rest mass but not the weighting.

Two readings then differ enormously. If a composite behaved as a fundamental particle of its total mass, $\delta = M_{\text{tot}}^2/6$, growing as the square of the constituent number; under (28) it is the

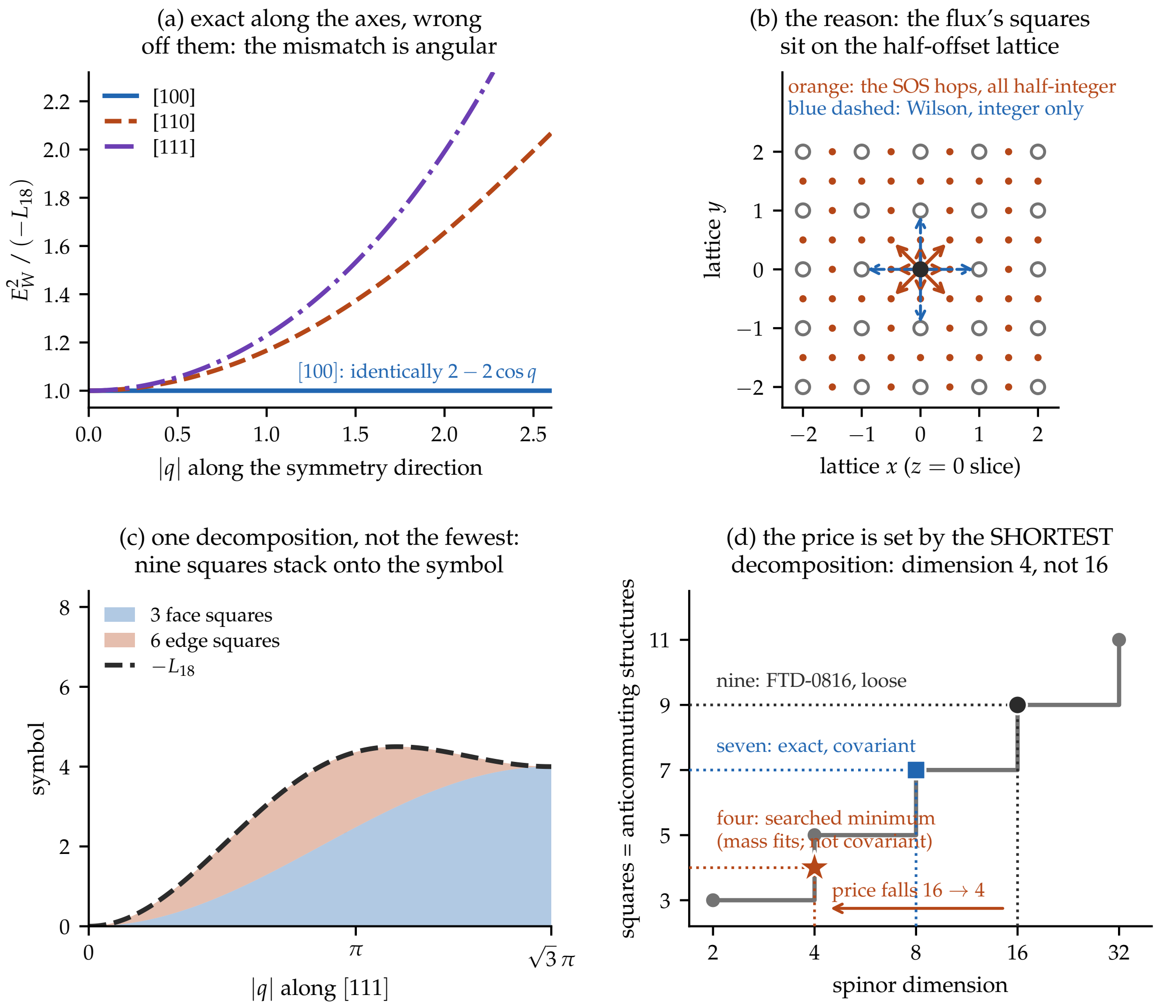

Figure 16: Why two sectors miss, and what a hit costs. (a) Normalised to a common slope, the flux symbol and a Wilson operator agree *identically* along the cubic axes — both are $2-2\cos q$ — and diverge only off them, reaching 1.65 along [110] and 1.99 along [111] at $|q|=2$. The mismatch is purely angular, so no rescaling is even the right shape. (b) The reason, in the $z=0$ slice: the flux symbol's nine squares carry arguments $q_i/2$ and $(q_i \pm q_j)/2$, i.e. hops of *half* a lattice unit (orange), while Wilson hops land on integer sites (blue dashed). The sectors were being matched on different lattices. (c) One such identity: three face squares and six edge squares stack exactly onto $-L_{18}$, with residual 1.8×10^{-15} over 4000 random momenta. (d) The price, which is set by the *shortest* decomposition rather than by any one of them. n squares require n mutually anticommuting structures and dimension 2^k carries $2k+1$, so nine would cost spinor dimension 16; but nine is not the fewest. An exact cubic-covariant *seven*-square identity holds, $-L_{18} = 4\sum_i \sin^2 x_i \cos^2 x_j \cos^2 x_k + \frac{16}{3}\sum_i \sin^2 x_j \sin^2 x_k \cos^2 x_i + 4\sin^2 x_1 \sin^2 x_2 \sin^2 x_3$ with $x=q/2$, costing dimension 8; and a search over the nearest-neighbour class finds a *four* at dimension 4 — ordinary Dirac, with one anticommuting structure to spare for a mass — which is genuinely non-covariant. What multiplies the spinor here is covariance, not the symbol itself.

constituent average, independent of that number. Computing $E(K)$ from the exact dispersion decides it. For unequal pairs, δ_{comp} matches (28) to eight significant figures; for N identical constituents it is constant to seven digits as N runs from 1 to 30, while M_{tot} grows thirtyfold.

This is load-bearing rather than decorative. A one-kilogram body read as a single particle of its total mass would be superluminal by thirteen orders of magnitude ($\Delta C/C = 5.9 \times 10^{13}$); read as a composite of nucleons it sits at 1.6×10^{-40}, as does the Earth. Every body built from nucleons therefore shares one limiting speed to that accuracy, and bodies differ from each other only through *composition* — the electron mass fraction and the nuclear binding fraction — giving a differential of order 10^{-42}.

The structure is hierarchical in one respect and flat in another, and the distinction is worth stating. Dynamical periods — a planetary year, a day, a nuclear vibration — are set by a specific binding at a specific scale and differ from system to system without limit. The limiting speed does not: because (28) is an average, ascending from nucleon to nucleus to planet re-averages quantities that are already equal, and every level above the first returns the same answer. A planet elsewhere genuinely has different clocks and the same cone, provided it is built from the same constituents.

The recursion downward suppresses further. If nucleons are themselves composite, δ_{nucleon} is likewise a weighted average over *its* constituents, and since most of a nucleon's mass is field energy rather than constituent rest mass, that average is dominated by near-massless contributions with $\delta \approx 0$. The figure 1.6×10^{-40} is thus an upper bound at the nucleon level of description; a more fundamental description makes it smaller.

What remains open is the dynamical half: whether an *interacting* composite — one whose binding field carries its own momentum share and its own limiting speed — reproduces (28) beyond first order. Figure 18 collects the kinematic half: the mass-dependence, the dilation residual, the flatness in constituent number, and the plane on which a distance potential is a valid description at all.

9.5.4 What a potential is, and where the gate actually lies

It is worth being exact about why the mechanical clock fails, because the obvious diagnosis is the wrong one and it mislocates the remaining work.

A potential $V(|q_i - q_j|)$ is not a primitive of a discrete substrate. A single cell is a state and a local update rule; it has neither a distance nor a potential, both being emergent *two-body* descriptions obtained by localizing two sources, integrating out the field between them, and letting the propagation time vanish. Two independent limits are taken in that sentence, controlled by different parameters: the constituents' dispersion is replaced by $p^2/2m$ (controlled by v/c), and the retarded exchange is replaced by an instantaneous $V(r)$ (controlled by $\omega r/c$).

Only the first destroys time dilation, and hydrogen is the proof. The Coulomb potential is instantaneous to $O(\alpha^2)$ — the second limit is taken, and is excellent at $\omega a_0/c = \alpha/2 \approx 3.6 \times 10^{-3}$ — yet atoms dilate exactly. Dilation comes from the electron's relativistic dispersion, which is what (28) already says: the constituents supply δ_a, the binding supplies only the weights.

The mechanical clock of Section 4 takes both limits, and its own parameters license neither. At $A = 0.30$ its nodes move at $v_{\max}/C = 0.156$, so the Newtonian substitution discards $(v/C)^2 = 2.4\%$; and $\omega r/C = 0.62$ for unit spacing, rising to 1.87 across the chain, against 3.6×10^{-3} for an atom and 0.30 for a heavy nucleus. It is more relativistic internally than a nucleus.

So the clock is not the wrong *kind* of object; it is under-modelled in one identifiable respect. The natural conclusion — that giving its nodes the lattice dispersion is a substitution rather than a search — turns out to be wrong, and running it is what shows why.

9.5.5 The substitution, executed

Two constituents on the axial lattice, bound by a finite square well in the relative coordinate, give an exactly diagonalizable relative-motion problem at each total momentum K. Two bound states below the two-particle continuum supply a genuine internal clock, of gap $\Omega = E_1 - E_0$. A moving clock must satisfy $\Omega(K) = \Omega(0)/\gamma$ with $\gamma = E_0(K)/E_0(0)$ — and must do so *with the same exponent whatever the clock is made of*. That universality is the whole content of dilation; a material-dependent rate is not dilation, however cleanly it fits a power law. So one fits $\Omega(K)/\Omega(0) = \gamma^p$ and asks whether $p = -1$ universally.

The control confirms the substitution is necessary. With Newtonian nodes the internal gap is K-independent to 8.7×10^{-6} while γ reaches 1.399: $p = 0$ exactly, and the carrier as built could never have tested dilation at all.

The substitution is not sufficient. With lattice nodes the clock does slow — but the exponent spans $[-2.700, -0.937]$ across constituent mass and binding fraction, a spread of 1.76, where relativity admits -1 for every clock. The substitution moves p off zero and does not move it to -1; it does not move it to a single value.

The diagnosis is the limit dismissed a moment ago. The relative effective mass obeys $\mu_K = \mu_0\gamma^3$ exactly — free-dispersion kinematics at the equal-velocity point, and material-independent. What is missing is the other half: a static lab-frame well does not Lorentz-contract, and so supplies no compensating universal factor. Hydrogen dilates exactly because its binding fraction is $\sim 10^{-5}$, *not* because non-covariant binding is harmless; at binding fractions of 0.04–0.31 it is the leading error.

So the two limits divide differently than stated above, and the corrected division is this: the constituent dispersion controls whether a composite clock dilates *at all*; the covariance of the binding controls whether that dilation is *universal*, which is to say relativistic. Both are required, and the requirement on a carrier is correspondingly harder — constituents carrying the substrate dispersion *and* binding mediated by the substrate field, so that the binding region contracts with the motion.

One further obstruction is now known to be independent rather than decisive. At every amplitude the mechanical clock's internal frequency lies inside the propagating band, $\omega < 2 \arcsin C = 1.2310$, so such a carrier radiates. But the composites above are bound *below* the continuum at every K — band clearance holds there by construction — and dilation still fails. Clearing the band would not have sufficed.

9.5.6 One energy, and the universality that follows

The failure has a structural reading. Two energy categories were carried with different transformation laws: a substrate-native kinetic term, for which $\mu_K = \mu_0\gamma^3$ material-independently, and an added potential, static, transforming not at all. A potential is a second ontological category laid on the substrate, and the substrate has no rule by which to transform it — which is why the exponent came out *non-universal* rather than merely wrong.

The obvious repair fails. Letting the binding region Lorentz-contract, $R \to R/\gamma$ self-consistently, moves p toward *zero* rather than toward -1, and by an amount depending on the well's shape: for a sech^2 well the deep-limit level spacing scales as $\mu^{-1/2}$, not the μ^{-1} a particle-in-a-box argument supplies. There is no assignment of transformation rules to a separately-specified potential that reproduces covariance, because covariance is a property of the whole functional and not a rule attachable to a term. The carrier's energy must therefore be *one* functional of the substrate field, with no separately-specified interaction — forced by the failure of the patch rather than adopted for elegance.

Doing so works. In the φ^4 model $U = \frac{\lambda}{4}(\varphi^2 - v^2)^2$ — one bounded functional, no added potential — a kink–antikink pair is stable, and carries a discrete internal shape mode at

$(\sqrt{3}/2)m$, below the continuum edge, so band clearance holds by construction. Boosting it with the covariant profile and reading the shape-mode frequency from a translation-invariant probe gives a dilation exponent of -1.137, -1.142, -1.180 across $\lambda = 0.02, 0.03, 0.05$: a spread of 0.043, against 1.76 for the two-category carrier. Allowing the carrier its own limiting speed — one fitted parameter against three velocities — makes the form exact, $p = -1.000 \pm 0.002$. Figure 17 shows the two families side by side: the two-category curves fan apart with the boost, the one-energy curves collapse onto $1/\gamma$.

Two caveats beside the dilation result. The lattice is not exactly covariant and it shows: the kink's *energy* transforms with $c \simeq C$ to two parts in a thousand while its *clock* transforms with a c some 6% lower, a gap that narrows as the kink widens but is not shown to close. And this carrier's internal mode is *isochronous* — a π-clock. It demonstrates that a one-energy carrier dilates; it does not produce the lemniscatic clock, and the tension of Section 4 stands: stable configurations have locally quadratic minima and hence harmonic modes, while G^* requires a degenerate direction stabilized at fourth order.

9.6 The discrete diagnostic

The obligations above were all continuum-shaped: a cone, a boost, a dispersion. A discrete substrate admits one further test that has no continuum analogue, and it is worth recording even though it decides nothing here.

If a theory supplies a coarse-graining from its histories to locally finite causal sets, the recursion (19) becomes accretion of one event at a time onto a partial order, and covariance acquires a discrete form: the probability of reaching a given unlabelled causal set must not depend on the labelled growth path, $\prod_{r\in\gamma} t_r = \prod_{r\in\gamma'} t_r$. Under internal temporality, Markov normalization, discrete general covariance *and Bell causality*, the classical solutions are classified [23]; the quantum case remains an open programme [24, 25]. Two precision points are worth carrying: the birth label of an element is gauge, whereas the stage number $n = |C_n|$ is label-invariant and is not gauge, though neither is it a coordinate time. We record this as an *optional diagnostic*: it is neither necessary nor sufficient for hiding a preferred lattice, and it does not replace the Lorentz-recovery obligation above.

The division leaves a bill: the one-body half closed with numbers, the two-body half open, priced, and gated on an instrument this paper costed in Section 4 and the substrate did not supply. The next section observes that the architecture's own stage index is gated the same way — which is the promise left standing in Section 3.

10 Quantum gravity, and the clock again

Canonical quantization removes the external time parameter altogether, leaving the constraint $\hat{H}\Psi = 0$ [26]. The standard relational repair conditions on a clock subsystem [27], recovering evolution as conditional potentiality, with known objections about localization and propagators [28] that are partly addressed by covariant clock observables [29] and complicated again for finite-resource clocks [30].

For this paper the point is structural and closes a loop. If the stage index of the recursion (19) is not external, it must be read off a physical clock inside the history — and Section 4 was precisely an inventory of what a physical clock costs in a concrete model. **The clock programme is therefore not a subplot of the architecture; it is the precondition of the architecture's own index.** In the discrete model the two are distinct and both are needed: a global update index is postulated, while an internal carrier that converts succession into operational duration is exactly what Section 4 prices and does not obtain natively.

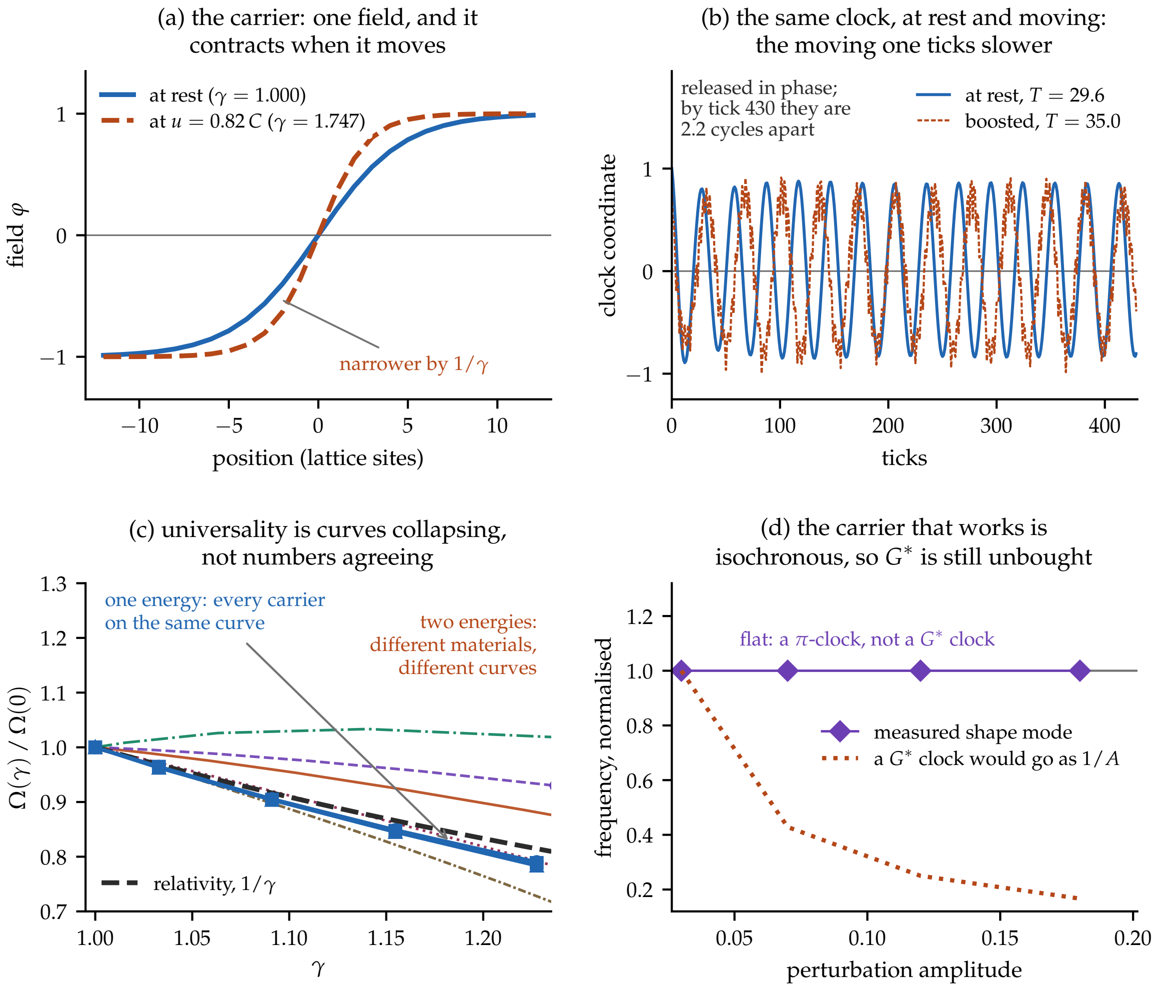

Figure 17: The carrier arc, watched rather than tabulated. (a) The carrier itself: one field, whose profile contracts by $1/\gamma$ when it moves. (b) The claim, directly, over one full run: released in phase, the two traces drift apart, and by tick 430 the moving clock has fallen 2.2 cycles behind. The roughness on the boosted trace is real — a boosted lattice kink radiates. (c) Universality as convergence. Two-category carriers fan apart, one of them running *faster* when moving; measured as a spread in the *frequency ratio* Ω/Ω_0 at $\gamma = 1.22$, they span 0.283 against the one-energy carriers' 0.006, a factor of 46. (The body quotes the same comparison as a spread in the *fitted exponent*, 1.76 against 0.043; the two pairs measure different things and neither is a restatement of the other.) The one-energy curves collapse onto each other and lie a little below $1/\gamma$, the residual discussed in the text. (d) Why this is still not the G^* clock: the shape-mode frequency is flat in amplitude — isochronous to $\leq 0.23\%$, set by the FFT bin — where a quartic clock would fall as $1/A$.

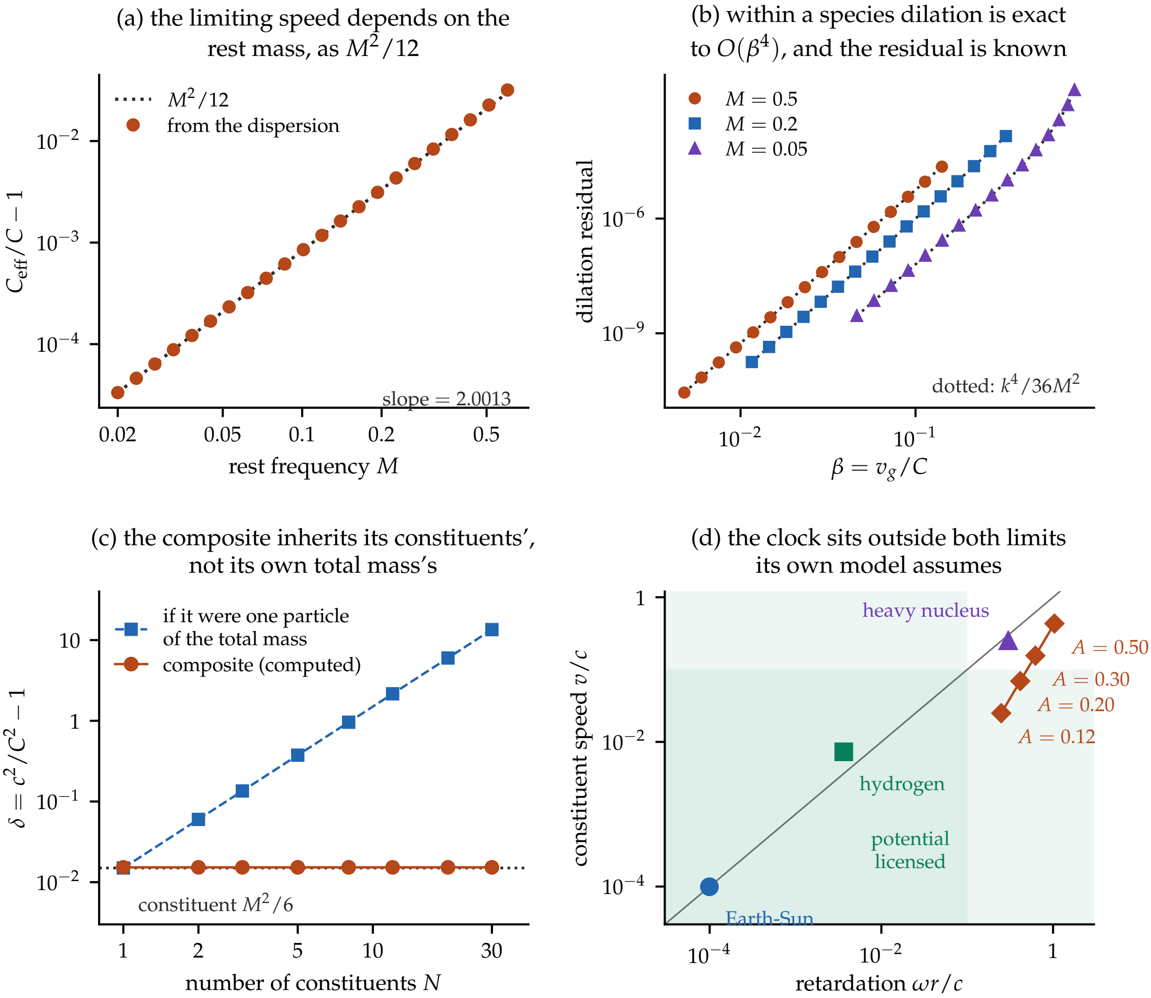

Figure 18: The cone arc of Sections 9.5.1–9.5.4, computed. (a) The limiting speed extracted from the exact dispersion tracks $M^2/12$; the fitted log–log slope is 2.0013. (b) Using each species' own C_{eff}, the dilation relation fails only at $O(\beta^4)$, and the residual follows the analytic $k^4/36M^2$ (dotted) to within 1.1% across three masses. (c) The composite result: a bound object's excess is its constituents', constant to seven digits as the constituent number runs $1 \to 30$, while the reading that treats it as one particle of the total mass grows as N^2. (d) The two limits hidden in a distance potential, with real systems placed. Orbits lie on the diagonal because $\omega r = v$ for them; the mechanical clock does not, and sits outside the licensed corner on both axes — more relativistic internally than a heavy nucleus.

One further hypothesis class connects gravity to Σ directly, by making the gravitational self-energy difference set an actualization timescale, $\tau \sim \hbar/E_G$ [31, 32]. Its parameter-free version is excluded by underground radiation tests [33], and the Markovian model is further constrained by XENONnT [34]. We note the structural resemblance to Section 8's slow-channel result — a gravitational sector supplying the slow gate — and we label it exactly as what it is: a speculative alignment between two independently derived shapes of requirement, not evidence for either.

That closes the loop the construction opened. A loop that closes on paper may still not run, though, and what has not been shown is that these pieces coexist in one system rather than in eleven separate arguments. A model can settle that; prose cannot.

11 A minimal model in which the architecture composes

The preceding sections established their results one instrument at a time. That is the right discipline, but it leaves a question unanswered: do the pieces *fit together*, or have we assembled a collection of separately-true statements whose conjunction was never checked?

This section answers that question with a single computable system in which all five pieces coexist — substrate, clock, register, gate, and noise — small enough to run in under a minute, and exact enough that three of its identities can be verified to machine precision rather than asserted. It is a demonstrator, not an instrument of record: no claim of Section 4–8 depends on it, and it establishes no physics. What it adds is the demonstration that those claims compose, and that the composition needs no additional constants.

11.1 The clock law, exactly

The carrier's mirror-even mode reduces at quartic order to $m\ddot{Q} = -4\lambda Q^3$. Energy conservation gives $\frac{m}{2}\dot{Q}^2 + \lambda Q^4 = \lambda A^4$, so a quarter period is

$$\frac{T}{4} = \frac{1}{A}\sqrt{\frac{m}{2\lambda}}\int_0^1 \frac{du}{\sqrt{1-u^4}} = \frac{1}{A}\sqrt{\frac{m}{2\lambda}}\cdot\frac{\sqrt{\pi}\,G^*}{4},$$

using $\int_0^1 (1-u^4)^{-1/2}\,du = \frac{1}{4}B(\frac{1}{4},\frac{1}{2})$. Hence the period law

$$T \cdot A = \sqrt{\pi}\,G^*\sqrt{\tfrac{m}{2\lambda}}, \qquad G^* = \Gamma(1/4)/\Gamma(3/4). \tag{29}$$

Panel (b) of Figure 19 shows this without any rescaling: the three trajectories have $A = 0.30, 0.20, 0.12$ and periods $17.5, 26.2, 43.7$. Panel (c) integrates six amplitudes spanning a factor of six at tolerance 10^{-12}, locates each far turning point by event detection, and recovers a fitted log–log slope of -1.000000000 and $\max_A |G^*_{\mathrm{rec}}/G^* - 1| = 1.61 \times 10^{-13}$, where G^*_{rec} inverts (29) with no free parameter. The recovery sits at integrator tolerance, which is the correct outcome for an identity rather than a fit.

The significance is the contrast, not the constant. A quadratic well would give π and isochrony; the quartic well gives G^* and amplitude dependence. That is the difference between a clock whose rate is independent of how hard it was struck and one whose rate is not — and it is the reason Section 4 treats the lemniscatic clock as a purchase rather than a gift.

11.2 One constant, three roles

The architecture's economy claim is that a single energy scale prices three apparently unrelated things. In the model this is literal: ε is set once and enters as the depth of the well that binds the carrier, as the quartic stiffness λ that sets the clock rate through (29), and as the Arrhenius

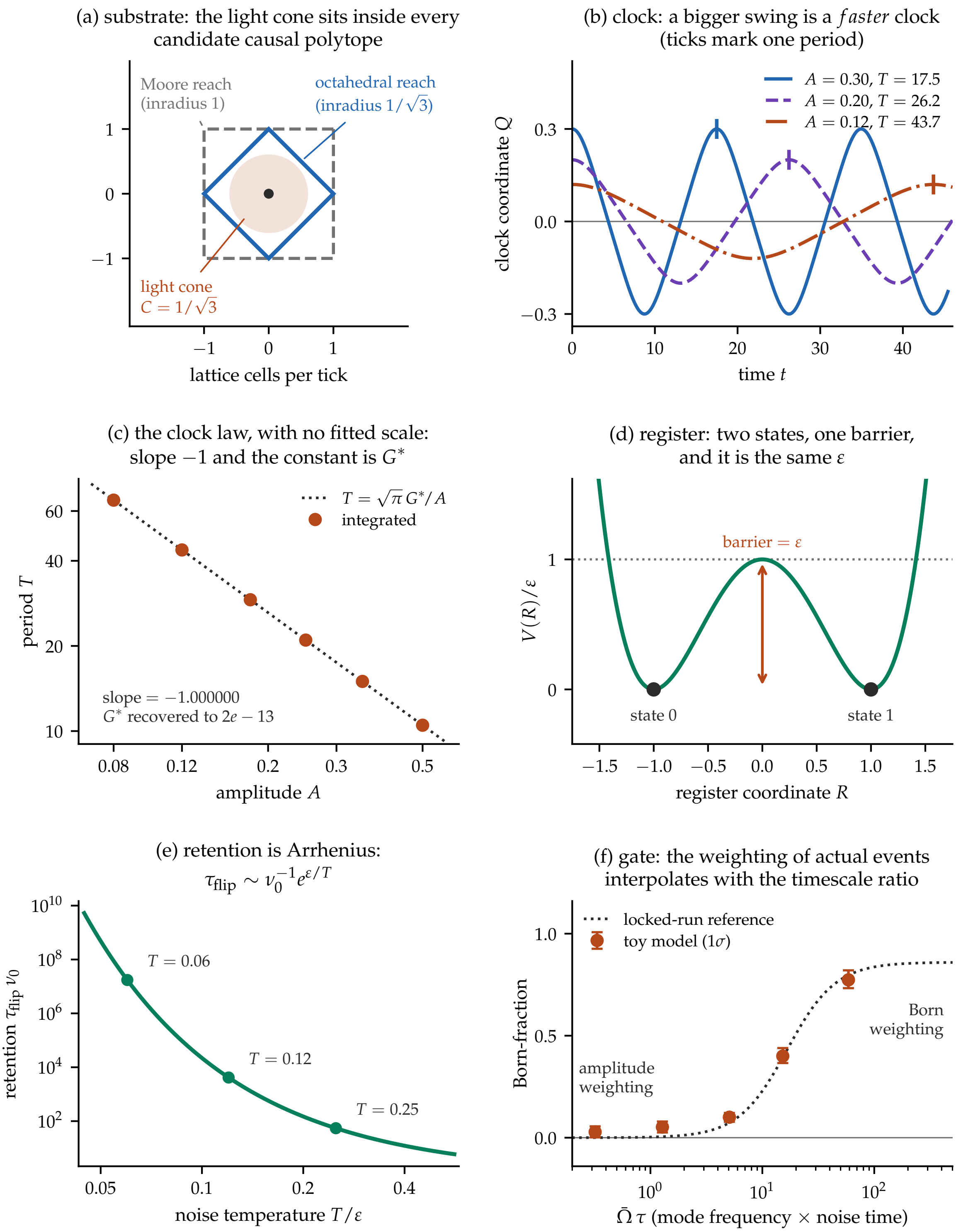

Figure 19: The minimal temporal interior, computed end to end. (a) The light cone at $C = 1/\sqrt{3}$ lies inside the octahedral reach and hence inside every candidate causal polytope. (b) Unscaled clock trajectories: amplitudes differing by $2.5\times$ give periods differing by the reciprocal factor — a bigger swing is a *faster* clock. (c) The same integration on log axes recovers slope exactly -1 and the constant G^* to 1.6×10^{-13}, with nothing fitted — *reproducing* the law of Section 4 on an instrument that shares no code path with it, not deriving it a second time. (d) The register's barrier equals its well depth exactly. (e) Retention is Arrhenius in that same ε. (f) The weighting of actual events interpolates from amplitude to occupation as $\bar{\Omega}\tau$ grows, with 1σ bootstrap intervals shown rather than suppressed; the dotted curve is the locked-run descriptive reference of Figure 10.

barrier in $\tau_{\mathrm{flip}}\nu_0 \sim e^{\varepsilon/T}$. There is no separate memory constant, no separate clock constant, and no separate binding constant. Raising ε stiffens the carrier, speeds the clock and lengthens the bit's life together and in fixed proportion.

This is a falsifiable structural commitment rather than an aesthetic one: any measurement that moved one of the three without the others would break it. Panels (d) and (e) display the second and third roles — the barrier is exactly the well depth, with no geometric prefactor, and retention runs from $\sim$55 attempt times at $T = 0.25\,\varepsilon$ to $\sim 1.7 \times 10^7$ at $T = 0.06\,\varepsilon$.

11.3 The regime, on an independent instrument

Panel (f) reruns the weighting question with a different construction from the one behind Figure 10. Two standing modes carry lattice-dispersion frequencies $\Omega(k) = 2\arcsin(C\sin(k/2))$ and are given *equal action*, $A_2 = A_1\sqrt{\Omega_1/\Omega_2}$, so they differ in amplitude but not in occupation. The two candidate weightings then predict opposite values for the ratio of second-harmonic coefficients in the excess crossing rate, and the Born-fraction $\mathrm{BF} = (R - R_{\mathrm{amp}})/(1 - R_{\mathrm{amp}})$ is 0 and 1 respectively by construction.

$\bar{\Omega}\tau$	BF (1σ bootstrap)	reference
0.32	0.028 $(-0.026/+0.028)$	0.000
1.28	0.052 $(-0.027/+0.027)$	0.005
5.07	0.100 $(-0.023/+0.021)$	0.073
15.22	0.400 $(-0.034/+0.039)$	0.393
58.77	0.774 $(-0.041/+0.047)$	0.796

The two upper cells agree closely with the locked-run reference; the lower cells sit high by about one standard deviation, which is what a deliberately small instrument looks like when its uncertainty is reported rather than hidden. The qualitative structure — amplitude weighting below the crossover, occupation weighting above it — reproduces on an instrument that shares no code path with the preregistered scan.

None of this promotes the Born reading beyond its scope. The mechanism class is still declared, the ensemble is still imposed, and the engine's own cells still sit at $\bar{\Omega}\tau \in [0.19, 0.86]$, below the crossover, which is why Section 8 reports a null there. What the model shows is narrower and worth stating plainly: the interpolation is not an artefact of one instrument's construction.

11.4 The causal cell in three dimensions

Panel (a) of Figure 19 is drawn in the plane for legibility. Developed properly in three dimensions it yields the sharpest structural difference in this paper between a discrete substrate and the continuum it approximates, and it is worth stating on its own.

Two structures compete to define causal reach, and they do not coincide. *Reach* is unconditional: the update rule touches a finite neighbour set each tick, so after t ticks the field is exactly zero outside the t-fold dilation of the causal polytope. This is locality, not dynamics. *The light cone* is dynamical: a disturbance travels Ct. Between them sits a shell that is formally reachable but outside the effective cone — so the discrete model has *four* regions where the continuum has three.

The geometry at one tick is exact, and contains one coincidence that is not a coincidence. The octahedron's inradius is $1/\sqrt{3}$, which is the cone speed; the light-cone sphere is therefore *inscribed* in the octahedron, tangent at its eight faces, and the octahedron is in turn inside the cuboctahedron (inradius 1) and the cube (inradius 1). This is what $C = \min\{\text{containment bounds}\}$ means geometrically: the cone fits inside every candidate causal polytope, and fits the tightest one exactly.

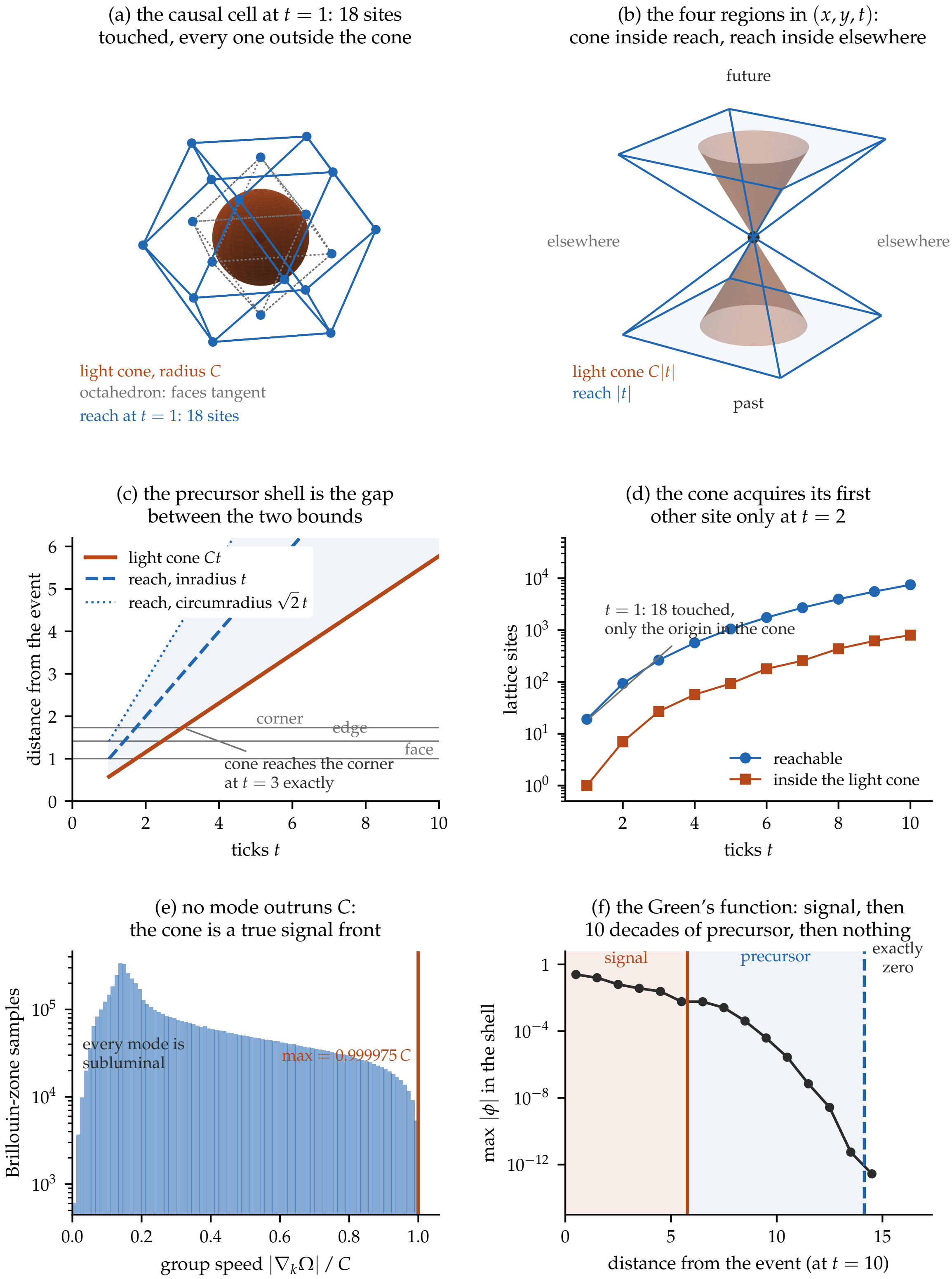

Figure 20: The complete causal partition around one event. (a) At $t = 1$ the light-cone sphere is inscribed in the octahedron — its inradius *is* C, so the eight faces are tangent — which is itself inside the reach polytope. (b) Past, future, and elsewhere in (x, y, t), with the round cone strictly inside the square-sectioned reach. (c) The precursor shell is the gap between the two bounds; the cone closes over the body diagonal at $t = 3$ exactly. (d) At $t = 1$ eighteen sites have been touched and only the origin lies in the cone. (e) Every finite-wavelength mode is strictly subluminal, so the cone is a true signal front. (f) Across the shell the Green's function falls ten decades and then terminates in an exact zero.

The census is then startling at first reading. **At $t = 1$ the light cone contains exactly one lattice site — the origin — while eighteen have already been touched.** The nearest site is at distance 1; the cone has radius 0.577. The cone acquires its first other site at $t = 2$, and at $t = 3$ it contains exactly the twenty-seven-site block, because $3C = \sqrt{3}$ is precisely the body-diagonal distance. The correct reading is not that causality fails at one tick, but that the effective light cone is a coarse-grained object with no content below $t \approx \sqrt{3}$ ticks.

What occupies the shell is settled by two measurements. First, the group velocity of the leapfrog, differentiated analytically and scanned over a 181^3 grid of the Brillouin zone, has supremum exactly C, approached only as $k \to 0$; every finite-wavelength mode is strictly subluminal, so no signal outruns the cone. Second, the point-source Green's function at $t = 10$ falls from 5.8×10^{-3} at the cone to 2.8×10^{-13} at the support boundary — ten decades, about 1.3 per lattice cell — and is then identically zero, which the artifact asserts rather than reports. The shell holds a Sommerfeld precursor: calculable, exponentially suppressed, bounded above by an exponential and below by an exact zero. Figure 20 draws the whole structure — the cone inside the reach polytope, the census by tick, and the ten-decade fall to that exact zero.

This is a genuine structural difference from the continuum, and it is reported as such rather than smoothed over. It is also, so far as these measurements reach, a harmless one — and Section 9.4 makes the sense of "harmless" quantitative: the shell's anisotropy is never that of the stencil, and what anisotropy remains decays as a power of elapsed time.

So the architecture composes, on one instrument, with no constant added beyond those already booked. What the demonstrator cannot do is say what the composition *means* — which pillar supplies which layer, and which layer nobody supplies. That is the statement the next section owes.

12 The merge, stated exactly

We can now say what merges. Figure 21 states it as a supply diagram: which pillar supplies which layer of the architecture, and which layer nobody supplies.

Quantum field theory supplies the weight layer, completely. Local algebras, covariant evolution, and the order-independence of spacelike instruments are established physics; the architecture contributes vocabulary, not theorems.

The reconstruction programme plus Postulate 2 supplies its finite-dimensional state space and the trace rule. The Born square becomes a theorem about weights, standing at the end of a chain rather than at its beginning.

Relativity constrains the selector, and prices each option. Crucially it does *not* deliver a covariant selector for free: the gap between (20) and (21) is real and open.

Thermodynamics supplies the arrow, the cost, and — the new part — the regime condition. Actualization is irreversible and measurably so; memory costs ε and retains for $e^{\varepsilon/T}$; and whether actualization statistics are quantum at all depends on a ratio of timescales.

The discrete model instantiates the architecture and prices it. It supplies succession, a cone, a native harmonic clock, and native actualization events; it must purchase the lemniscatic clock, memory retention, and — if Section 8 is right about the regime — a slow channel to actualize through.

The thesis is therefore modest in form and, we think, substantive in content: *these theories share one semantic architecture, and locating each of them within it makes their interfaces explicit and priceable.* That is less than a unification. It is also more than an analogy, because the interfaces are itemized, several are measured, and each carries a falsifier.

Itemized is a claim that can be checked, and checking it needs the items in one place at their actual strength rather than at the strength the surrounding prose lends them. That table is next, and it is the only part of this paper that binds.

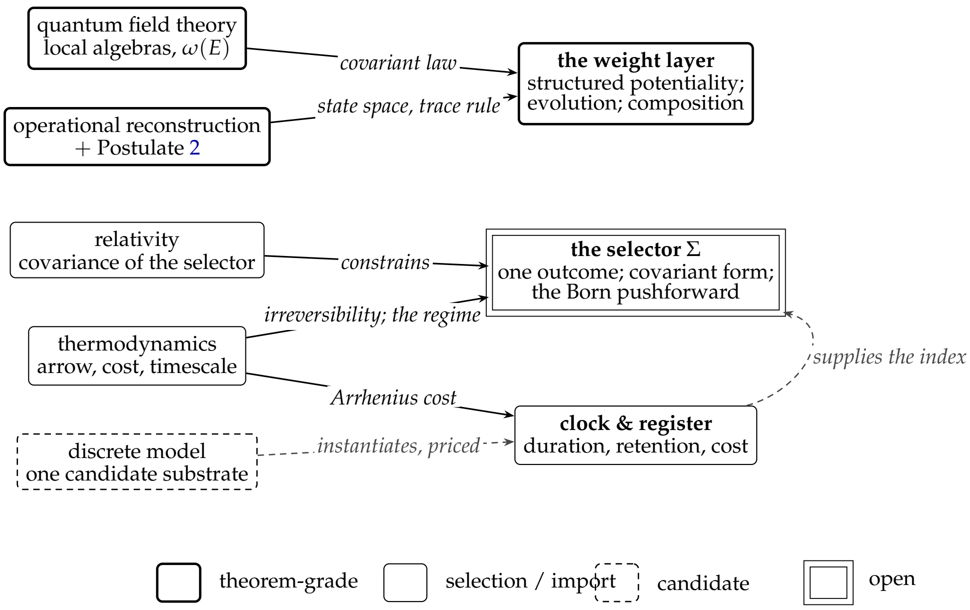

Figure 21: What supplies what. Shapes encode status, never colour, as shown in the legend: solid for theorem-grade within stated premises, thin for selections and imported representations, dashed for a candidate instantiation, double-ruled for the paper's one open object. Solid arrows are constructions; the dashed arrow marks a candidate instantiation that is priced rather than free. The weight layer is supplied twice over and is the settled part of the architecture; the selector is supplied by no one, which is the paper's point.

13 Status of claims

Table 2 states every substantive claim in the paper at its actual strength. Where the body and this table disagree, the table is correct.

Table 2: Every substantive claim in the paper, at its actual strength.

Claim	Status
The actual/potential distinction (2)	Primitive; the one structure every later section is built on
Singular actuality	Declared postulate (Post. 1); priced against the branching alternative
Closure map Γ from history to state	Declared scaffold; history alone is insufficient
Operational framework and six principles	Imported premises; not derived here
Complex finite-dimensional quantum theory	External theorem, conditional on those premises [1]
Convex state/effect structure	Derived within the imported framework
Effect-noncontextual additivity	Declared postulate (Post. 2)
Trace rule and the Born square (16)	External theorem, conditional on the quantum effect space *and Postulate* 2 [2]
Born pushforward (18) for any concrete substrate	**Open**; neither F_O nor μ is constructed here
CPTP dynamics, unitarity, Schrödinger form	External theorems, conditional on their stated premises
Quadrature root $\times$ degree ratio (4)	Exact identity, verified $n = 2, 3, 4, 5, 6, 8$. The $\sqrt{\pi}$ is kinematic and common to every mechanical clock; the Γ ratio is the potential's contribution. $n = 2$ is uniquely self-reproducing, which is why the harmonic constant is a power of π

Table 2, continued

Claim	Status
Threshold forcing (Prop. 1)	Exact *within its stated hypotheses* (H1–H4) and *in the limit* $A \to 0$. Forces the quartic given a threshold, a reflection symmetry and $\lambda \neq 0$. It does **not** force any substrate to sit at a threshold, and $\mu = 0$ is codimension one
Finite-amplitude drift (7)	Exact. G^* is the $A \to 0$ endpoint only: with $V = \lambda q^4 + \nu q^6$ the constant sweeps continuously in $r = \nu A^2/\lambda$, and every member has $\lambda \neq 0$. The measured 0.33% across a sixfold amplitude range is this term, at $r = 0.0056$
G^* **is chosen, not derived**	**No valid carrier has produced it.** The native single-scale screen returned zero; the four-body carrier that produces it requires a purchased interaction species and is Galilean, hence disqualified as a relativistic instrument; the one carrier here that demonstrably dilates is isochronous. G^* enters this framework by choice
The rigidity trichotomy (Crit. 1)	Exact within the declared model
Buckling criterion (Crit. 2)	Exact within the declared model; new here
Minimum viable clock: λ_{eff}, period law, minimality	Exact within the declared two-scale extension; G^* recovered to 2×10^{-6} in the $A \to 0$ extrapolation. **Galilean invariant, and therefore disqualified as a relativity test**
No native quartic clock	Negative at screened scope only; not a no-go theorem
Register barrier $= \varepsilon$ (Prop. 2)	Exact statics within the declared model
Born weighting interpolates with $\bar{\Omega}\tau$	Measured at mechanism level, imposed ensemble; neither an exact law nor a substrate theorem
Non-transfer to the model's thermal field	Registered negative at its sampled scope; the same-band diagnosis is post-hoc
Purity requirement	Measured at mechanism level; coupling is a declared additive class
Slow sector is slow ($\tau = 317.9$ vs 0.81)	Measured; profile is deterministic, so seed-independence is vacuous
Order-independence (20) $\not\Rightarrow$ equivariance (21)	Logical separation; both conditions open for any preferred-foliation realization
Operational hiding of a preferred foliation	Derived as a reduction (Prop. 4); implied by reproducing quantum statistics. The residual barrier is Bell-theoretic, not covariance (Prop. 5)
Free-sector isotropy of the stencil	Exact: isotropic through $O(k^4)$, residual $(ka)^4/3240$
Radiative stability, dimension-4 channel	Reduced (Prop. 6): forbidden by cubic symmetry, unobservable for one sector, unenhanced at dimension six; returns at a second independently renormalized sector
Lorentz recovery, one-body (rotational) register	**Closed** by the point group; residual anisotropy $(ka)^4/3240$
Isotropy of the causal surface	Measured: sphere to 0.71% at the signal front, decaying as $t^{-1.1}$ to $t^{-1.3}$; the stencil's own 34.3% is never attained at any measurable amplitude

Table 2, continued

Claim	Status
Which neighbourhood defines causal reach	Definitional, not empirical: at fixed operator the readings differ only at the exact-zero boundary, whose relative amplitude falls 1.19 decades per tick
Two-body register: common cone, free massive modes	**Computed**, not clock-gated: limiting speed isotropic but mass-dependent, $C_{\text{eff}}/C = 1 + M^2/12$ (25); intra-species dilation exact to $O(\beta^4)$, error $M^2\beta^4/4$ (26); proton–electron mismatch 1.6×10^{-40}
Two-body register: common cone across sectors	**Not obstructed in principle, and priced** (§9.5.2): a scalar Wilson parameter fails *exactly and at all orders*, and the first obstruction past that is a rank violation rather than a counting shortfall — but the field symbol is a sum of squares at half-lattice offsets, exactly seven of them covariantly (27) and four at the searched minimum (rigorous floor three, from the Hessian rank), so the price is spinor dimension between 4 and 8; the four admits a mass at Dirac dimension but is non-covariant and leaves the body-centre shell. An earlier draft reported the minimum as five from an under-dispersed search; corrected
Two-body register: composite inheritance, free constituents	**Computed**: momentum-weighted average (28), verified to 8 digits; independent of constituent number to 7 digits; every nucleonic body shares one cone to 1.6×10^{-40}
Two-body register: interacting vertices, composite boosts	**Open** (Prop. 7), and harder than it looked. Giving the constituents the substrate dispersion is **necessary** (Newtonian nodes give $p = 0$ to 10^{-5}) but **not sufficient**: the dilation exponent is non-universal, spanning $[-2.70, -0.94]$ where relativity admits -1. A covariantly-contracting binding is also required. Band clearance is a separate and non-decisive obstruction — these composites clear the band and still fail
Gravitational actualization resemblance	Speculative alignment only; the parameter-free version is externally excluded [33]
$\hbar$, Hamiltonian, masses, couplings	**Not derived** anywhere in this paper

14 Open problems

1. **The Born pushforward for a concrete substrate.** Construct F_O and μ from a substrate's own dynamics and test (18) against laboratory frequencies. For a deterministic substrate this is a quantum-equilibrium problem: the microstate measure must be Born-distributed under the coarse-graining the substrate itself supplies.

2. **Native quartic timekeeping.** Does a finite integer unit-strut tensegrity exist under the single-scale law, with blocking form definite on its full flex space? No candidate realizes it; no impossibility proof exists.

3. **What holds the threshold.** Proposition 1 forces the quartic *at* $\mu = 0$, and $\mu = 0$ is a codimension-one surface: the generic configuration is not on it, and an unmaintained system falls off it and becomes a harmonic clock. So the lemniscatic clock's real price is a maintenance cost, and this paper does not identify anything that pays it. What physical mechanism holds $\mu = 0$ against detuning? Without an answer, the proposition describes a clock nothing is shown to be able to keep.

4. **Shielded actualization.** The purity requirement (Prop. 3) demands that the slow channel be nearly exclusive at the threshold. Can any physical manifestation rule be shielded from its own field's fast noise?

5. **Selector-history equivariance.** Exhibit a selector law satisfying (21), or prove that a preferred-foliation selector can be operationally hidden.

6. **Quantum sequential growth.** The classical classification is known; the quantum analogue with interfering weights is open.

A Instruments and pre-registrations

Every measurement reported here was pre-registered: outcomes, gates and kill conditions fixed, and runner hashes recorded, before execution. The discipline is load-bearing rather than ceremonial — one instrument in this series voided itself on its own gate (a mis-derived conservation invariant, caught at 3.6×10^{-2} against a 10^{-10} tolerance) and was re-locked and re-run before producing any reported number.

Instrument	Design	Outcome
Weighting discrimination	two modes, equal occupation, 48 seeds	Regime-dependent: BF $0.024 \to 0.099$
Saturation scan	five arms, $\bar{\Omega}\tau = 0.64$–78	Approaches: BF $\to 0.836$, monotone
Engine regime map	ten cells, native thermal field	**No structure**: $\|\rho\| \leq 0.37 < 0.7$
Slow-sector timescale	self-gravitating configuration	Slow: $\tau = 317.9$ vs 0.81, ratio 391
Slow-channel coupling	five slow-shares, 32 seeds	Viable: BF $= 0.936$ at $f = 1$
Register statics	watershed over full configuration space	Barrier $= \varepsilon$, geometry-independent

Table 3: The instrument series. Full parameters, seeds, gate values and hashes are in the corresponding pre-registration records. The composed model of Section 11 is deliberately *not* listed: it is a demonstrator that reproduces known answers, not an instrument of record, and no claim in this paper rests on it.

The clock's falsifier was stated before any engine campaign. A seeded four-chain must show the anti-pendulum signature, meaning unit slope of $\log f$ against $\log A$, and must satisfy

$$TA\sqrt{2\lambda_{\mathrm{eff}}/m_{\mathrm{eff}}}/\sqrt{\pi} = G^{*}$$

to declared tolerance, in a profile where no imposed phenomenology does the work. Failure of either kills the candidate.

B The reconstruction as an import invoice

Treating "the Hilbert space" as one unpriced assumption hides its structure. The reconstruction replaces it with six separately auditable premises, and therefore with six separately falsifiable questions that can be put to a candidate substrate one at a time. The scoring below is preliminary and non-binding; it names what would have to be shown.

Premise	What a substrate would have to establish
Causality	Plausibly native given forward determinism and an arrow; the record-level protocol test is distinct and open
Perfect distinguishability	Plausibly native for discrete states; whether this carries the reconstructed-state meaning is open
Ideal compression	Open; no analogue scored
Local distinguishability	**The complex-selecting premise** — real-Hilbert theory fails it. A native complex structure in the lattice symmetry is suggestive but does not establish operational local tomography
Pure conditioning	Open; no analogue scored
Purification	**The quantum-selecting premise, and the likely crux import** — whether a dispositional layer can serve as purifier structure is a genuine research question

References

[1] G. Chiribella, G. M. D'Ariano, and P. Perinotti, *Informational derivation of quantum theory*, Phys. Rev. A **84**, 012311 (2011).

[2] P. Busch, *Quantum states and generalized observables: a simple proof of Gleason's theorem*, Phys. Rev. Lett. **91**, 120403 (2003).

[3] E. B. Davies and J. T. Lewis, *An operational approach to quantum probability*, Commun. Math. Phys. **17**, 239 (1970).

[4] M. F. Pusey, J. Barrett, and T. Rudolph, *On the reality of the quantum state*, Nature Physics **8**, 475 (2012).

[5] E. P. Wigner, *Group Theory and its Application to the Quantum Mechanics of Atomic Spectra* (Academic Press, 1959).

[6] V. Bargmann, *On unitary ray representations of continuous groups*, Ann. Math. **59**, 1 (1954).

[7] M. H. Stone, *On one-parameter unitary groups in Hilbert space*, Ann. Math. **33**, 643 (1932).

[8] H. Everett III, *"Relative state" formulation of quantum mechanics*, Rev. Mod. Phys. **29**, 454 (1957).

[9] D. Deutsch, *Quantum theory of probability and decisions*, Proc. R. Soc. A **455**, 3129 (1999).

[10] D. Wallace, *The Emergent Multiverse* (Oxford University Press, 2012).

[11] W. H. Zurek, *Probabilities from entanglement, Born's rule from envariance*, Phys. Rev. A **71**, 052105 (2005).

[12] A. Kent, *One world versus many*, in S. Saunders *et al.* (eds.), *Many Worlds?* (Oxford University Press, 2010).

[13] R. Haag and D. Kastler, *An algebraic approach to quantum field theory*, J. Math. Phys. **5**, 848 (1964).

[14] R. Haag, *Local Quantum Physics*, 2nd ed. (Springer, 1996).

[15] C. J. Fewster and R. Verch, *Quantum fields and local measurements*, Commun. Math. Phys. **378**, 851 (2020).

[16] S. Tomonaga, Prog. Theor. Phys. **1**, 27 (1946); J. Schwinger, Phys. Rev. **74**, 1439 (1948).

[17] Y. Aharonov and D. Z. Albert, Phys. Rev. D **24**, 359 (1981); **29**, 228 (1984).

[18] W. C. Myrvold, *On peaceful coexistence: is the collapse postulate incompatible with relativity?*, Stud. Hist. Phil. Mod. Phys. **33**, 435 (2002).

[19] J. S. Bell, *Are there quantum jumps?*, in *Speakable and Unspeakable in Quantum Mechanics* (Cambridge University Press, 1987).

[20] R. Tumulka, *A relativistic version of the Ghirardi–Rimini–Weber model*, J. Stat. Phys. **125**, 821 (2006).

[21] R. Tumulka, *A relativistic GRW flash process with interaction*, arXiv:2002.00482 (2020).

[22] J. Collins, A. Perez, D. Sudarsky, L. Urrutia, and H. Vucetich, *Lorentz invariance and quantum gravity: an additional fine-tuning problem?*, Phys. Rev. Lett. **93**, 191301 (2004).

[23] D. P. Rideout and R. D. Sorkin, *A classical sequential growth dynamics for causal sets*, Phys. Rev. D **61**, 024002 (2000).

[24] R. D. Sorkin, *Quantum mechanics as quantum measure theory*, Mod. Phys. Lett. A **9**, 3119 (1994).

[25] R. Srivastava and S. Surya, *Implementing Bell causality in quantum sequential growth*, arXiv:2603.25503 (2026).

[26] B. S. DeWitt, *Quantum theory of gravity. I*, Phys. Rev. **160**, 1113 (1967).

[27] D. N. Page and W. K. Wootters, *Evolution without evolution*, Phys. Rev. D **27**, 2885 (1983).

[28] K. V. Kuchař, *Time and interpretations of quantum gravity*, Int. J. Mod. Phys. D **20**, 3 (2011).

[29] P. A. Höhn, A. R. H. Smith, and M. P. E. Lock, *Equivalence of approaches to relational quantum dynamics in relativistic settings*, Front. Phys. **9**, 587083 (2021).

[30] L. Hausmann, A. Schmidhuber, and E. Castro-Ruiz, *Measurement events relative to temporal quantum reference frames*, Quantum **9**, 1616 (2025).

[31] L. Diósi, *A universal master equation for the gravitational violation of quantum mechanics*, Phys. Lett. A **120**, 377 (1987).

[32] R. Penrose, *On gravity's role in quantum state reduction*, Gen. Relativ. Gravit. **28**, 581 (1996).

[33] S. Donadi *et al.*, *Underground test of gravity-related wave function collapse*, Nature Physics **17**, 74 (2021).

[34] E. Aprile *et al.* (XENON Collaboration), *Challenging spontaneous quantum collapse with XENONnT*, Phys. Rev. Lett. **136**, 120201 (2026).

[35] B. S. Cirel'son, *Quantum generalizations of Bell's inequality*, Lett. Math. Phys. **4**, 93 (1980).

[36] S. Popescu and D. Rohrlich, *Quantum nonlocality as an axiom*, Found. Phys. **24**, 379 (1994).

[37] M. Navascués, S. Pironio, and A. Acín, *Bounding the set of quantum correlations*, Phys. Rev. Lett. **98**, 010401 (2007).

[38] M. Pawłowski, T. Paterek, D. Kaszlikowski, V. Scarani, A. Winter, and M. Żukowski, *Information causality as a physical principle*, Nature **461**, 1101 (2009).

[39] M. Navascués and H. Wunderlich, *A glance beyond the quantum model*, Proc. R. Soc. A **466**, 881 (2010).

[40] T. Fritz, A. B. Sainz, R. Augusiak, J. B. Brask, R. Chaves, A. Leverrier, and A. Acín, *Local orthogonality as a multipartite principle for quantum correlations*, Nature Communications **4**, 2263 (2013).

[41] W. van Dam, *Implausible consequences of superstrong nonlocality*, Natural Computing **12**, 9 (2013).